
Mechanistic and Statistical Modelling of Bioprocesses

This chapter provides a general introduction for the application of mechanistic and statistic models in bioreactor process development and optimisation. We first introduce the equations governing the behaviour of batch and continuous stirred tank reactors (CSTR) for a given chemical system. After presenting the criteria to generalise these concepts to biological systems and bioprocesses, we discuss the implementation of mechanistic models for the simulation and control of the bioreactor performance with particular emphasis on product quality attributes, such as N-linked glycosylation. In addition, we describe the implementation of statistical and hybrid models and their application in process development and reactor optimisation. Lastly, we compare the use of the various modelling techniques for process monitoring and control.

6.1 ROLE OF MATHEMATICAL MODELS IN UPSTREAM PROCESSES

In the previous chapters, we have introduced and discussed the most relevant concepts and hardware requirements in order to design, develop, optimise, and scale up perfusion cell culture processes. In this chapter, we discuss the use of mathematical modelling for upstream processes. Mathematical models remain the primary tool for the quantitative understanding of processes, and their use can fundamentally improve our confidence in the way we design and operate them. As such, models provide a solid basis for changing from the current state-of-the-art recipe production towards real-time online monitoring and closed-loop production control, as well as towards a better understanding of the sources of variability in processes and products (Rathore 2009, 2014, Undey et al. 2011). The applications of numerical simulations are manifold. Models can be used at the process design and development level by helping the search for better operating conditions (Karst et al. 2017b), the design of an experimental campaign intended to reach a specific objective (Brühlmann et al. 2017b, Narayanan et al. 2019c, Sokolov et al. 2018) and support the operation and control of a running unit (Steinebach et al. 2016a). The latter can be accomplished by introducing elements of forward control action or, in a more structured approach, as part of model predictive control strategies (Morari & Zafiriou 1989). Mathematical models can also be extremely valuable for the analysis of the potential of a given technology, or even at the level of process characterisation and validation for regulatory purposes (Baur et al. 2015). Models also represent a core component of several initiatives launched by regulatory agencies, such as process analytical technology (PAT) and the quality by design (QbD).

6.2 Modelling of Chemical Systems

Despite recent developments in understanding the cell machinery and its interactions with the culture environment, process modelling in the biopharmaceutical industry is still at its infancy compared to other fields, given the enormous complexity of living cells (Jagschies et al. 2018). Currently, bioprocess modelling is used to simulate bioreactors at the macroscopic scale and specific biological processes at the cell level to tune the quality characteristics of the final product. Bioreactor models are based on the description of the kinetics of the chemical reactions, the mass transport among the different phases and the fluid-dynamic behaviour of the suspension inside the bioreactor. From the cell biology side, cellular processes have to be described to a suitable level of detail – such as growth, metabolism, protein production and modification pathways on a single-cell or cell-population level. These descriptions can be based on first principles, like mass or energy balances, thus leading to mechanistic models, or the elaboration of available historical data to draw correlations of statistical nature (Kiparissides et al. 2011, Kontoravdi et al. 2010). Both these types of models not only play a significant role in process understanding but may also be useful for the implementation of soft sensors and process control (Carrondo et al. 2012, Luttmann et al. 2012). For example, the spectra collected by Raman or NIR sensors in a bioreactor need to be processed and correlated to certain observed variables in order to build predictive models that can be used to implement such sensors for process monitoring, optimisation and control.

In the next sections, we are going to introduce the basic concepts for the modelling of chemical systems. We will then provide the criteria and assumptions for applying this framework to biological systems with respect to different bioreactor operating modes.

6.2 MODELLING OF CHEMICAL SYSTEMS

Let us introduce the concept of reactor modelling in very general terms in order to understand which kind of knowledge is needed to build such models and which kind of information about the reactor behaviour we can extract from them. In particular, we refer to stirred tank reactors, where we assume ideal mixing conditions so that the reactor content is uniform (i.e., there is no difference in composition or temperature between any two points inside the reactor). The mass balance in such a reactor can be written in the following general form (Fogler 2008):

$$\frac{dN_i}{dt} = F_i^{in} - F_i + G_i, \quad (6.1)$$

where $\frac{dN_i}{dt}$ is the rate of accumulation of species i inside the reactor, N_i is the number of moles of the species i inside the reactor, F_i^{in} is the molar flowrate entering the reactor, F_i is the molar flowrate leaving the reactor, and G_i describes the rate of production of species i through any chemical or biological transformation. Introducing c_i , the molar concentration of species i uniform inside the reactor, and thus also characterising the outlet stream, we can use a simplified form of Equation (6.1):

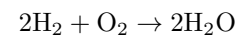
$$\frac{d(c_i V_R)}{dt} = Q_{in} c_{i,in} - Q_i c_i + r_i V_R, \quad (6.2)$$

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with Q representing the volumetric flowrate and r_i representing the rate of production of species i , defined by

$$r_i = \frac{\text{moles of } i \text{ produced}}{\text{time} \times \text{volume}}. \quad (6.3)$$

In addition, as we are dealing with molar species, we have to consider the stoichiometry of any given reaction – which, for the simple case of the synthesis of water out of oxygen and hydrogen, is given by



where v_i represents the stoichiometry coefficient for species i , and it is negative if the species is consumed in the reaction and positive otherwise. Due to stoichiometry, the ratio of the rate of production for each species i and the corresponding stoichiometric coefficient v_i is the same

$$\frac{r_i}{v_i} = \frac{r_{\text{H}_2}}{-2} = \frac{r_{\text{O}_2}}{-1} = \frac{r_{\text{H}_2\text{O}}}{2} = R, \quad (6.4)$$

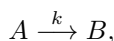
where R represents the overall rate of the reaction. In the case that several reactions take place at the same time, the overall rate of production of a given species i , r_i is given by the sum of the contributions of all different reactions, as follows:

$$r_i = \sum_{j=1}^{N_R} v_{i,j} R_j \quad i = 1, 2, \dots, N_C \quad (6.5)$$

where N_R represents the number of reactions, N_C the number of species, $v_{i,j}$ the stoichiometric coefficient of species i in reaction j , and R_j the rate of reaction j .

In order to define a mechanistic model for the kinetics of a given reaction, we can proceed as follows.

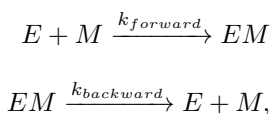
For a simple reaction describing the irreversible transformation of A into B,



we can derive the following kinetic model, assuming first order with respect to the reactant:

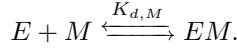
$$r = k c_A, \quad (6.6)$$

where k represents the reaction rate constant and typically exhibits a temperature dependence, as described by the Arrhenius law. If we now consider the slightly more complex case of the reversible reaction between a protein enzyme E and a metal cofactor M to form the activated metal enzyme complex EM and its dissociation:



6.3 Modelling of Biological Systems

we can derive an equation describing the reaction kinetics by first introducing the equilibrium constant $K_{d,M} = \frac{k_{forward}}{k_{backward}}$ representing the equilibrium conditions for the activated metal enzyme complex:



Next, we introduce the following expressions of the relevant reaction rates:

$$r_{forward} = k_{forward}C_{ECM} \quad (6.7)$$

$$r_{backward} = k_{backward}C_{EM} \quad (6.8)$$

$$R = r_{forward} - r_{backward} \quad (6.9)$$

$$R = k_{forward}C_{ECM} - \frac{k_{forward}}{K_{d,M}}C_{EM}, \quad (6.10)$$

where R represents the overall net reaction rate, which at equilibrium conditions is zero, thus leading to

$$k_{forward}C_{ECM} = k_{backward}C_{EM}, \quad (6.11)$$

which leads to the classical definition of the reaction equilibrium constant for the case of association of the enzyme and the metal:

$$\frac{k_{forward}}{k_{backward}} = K_{d,M} = \frac{C_{ECM}}{C_{EM}}. \quad (6.12)$$

The description of the reaction mechanism and the derivation of the corresponding expression for the reaction rate becomes more complicated for larger systems, but it can still be accomplished using the procedures typical of chemical reaction engineering (Fogler 2008, Froment et al. 1990).

6.3 MODELLING OF BIOLOGICAL SYSTEMS

The concepts developed above for the description of chemical systems can be applied to biological processes, since in principle any fermentation can be considered as the transformation of a given mixture of reactants (metabolites) into a mixture of products (biomass, proteins, lipids, sugar and other organic molecules). In principle, two categories of mechanistic models for cell culture modelling are considered: structured and unstructured (Bibila & Robinson 1995). While unstructured models neglect the inner structure of the cell, the structured ones try to account for it by considering the cell as divided in distinct compartments, where different biochemical reactions are taking place (Kontoravdi et al. 2010, Pörtner & Schäfer 1996). It should be noted that even the most detailed and biologically consistent simulation of single-cell behaviour often fails to describe the collective behaviour of an entire population of cells (Sidoli et al. 2006). Unstructured models, on the other hand, are developed to provide a first description of the entire population and, representing an empirical basis for applying control, optimisation and process development techniques (Jang & Barford 2000). Unstructured models rely on extracellular culture variables, which are the only ones typically monitored during cultivations. This translates in a reliability domain that is usually limited to the range of experimental conditions from which they have

been derived, which is often small compared to the large space of process design and optimisation. However, Pörtner & Schäfer (1996) indicated that certain cellular characteristics, such as the growth of different cell lines, follows similar kinetic rules, irrespective of the cultivation mode. Here, we present the basic concepts for unstructured cell culture modelling in bioprocessing, which is currently the most widely used approach.

The main variable for fermentation processes is the total viable cell density (or biomass) within the system. Living cells reproduce through mitosis – i.e., one organism gives rise to two exact copies of itself. If the environmental conditions are favorable, the number of cells grows exponentially in time with a specific growth rate μ [time⁻¹]. For the viable cell density X_V , the corresponding mass balance takes the form:

$$\frac{dX_V}{dt} = \mu X_V. \quad (6.13)$$

The growth rate μ is usually described as a function of the extracellular environment. For example, growth can be inhibited by reducing the temperature of the system (Wolf et al. 2019a) and largely depends on the concentration of metabolites and growth inhibitors. This effect can be modelled by considering additional terms in the calculation of the growth rate (Sou et al. 2017):

$$\mu = \mu_{max} f_{lim} f_{inh}, \quad (6.14)$$

where μ_{max} is the maximum cell growth rate of the cells at optimal conditions (i.e., maximal nutrient availability and no inhibitions), while f_{lim} is the term indicating nutrient limitation, which is usually modelled as Monod equation with respect to the metabolites of interests $c_1, \dots, c_n$:

$$f_{lim} = \frac{c_1}{K_1 + c_1} \frac{c_2}{K_2 + c_2} \cdots \frac{c_n}{K_n + c_n}, \quad (6.15)$$

where K_n is the Monod constant for the limiting metabolite c_n . For mammalian cell cultures, the medium components that represent the term c_n are usually glucose, particular amino acids and growth factors. The term f_{inh} can be written as

$$f_{inh} = \frac{K_1}{K_1 + I_1} \frac{K_2}{K_2 + I_2} \cdots \frac{K_n}{K_n + I_n}, \quad (6.16)$$

where I_n indicates the concentration of a given growth inhibitor n , such as lactic acid or ammonia. The concentrations of metabolites and inhibitors follows a general mass balance:

$$\frac{dc_i}{dt} = q_i X_V + Q_{in} c_{i,in} - Q_{out} c_i, \quad (6.17)$$

where q_i indicates the specific production rate of the species i and the terms Q_{in} and Q_{out} are the volumetric flowrates that account for feeding or removal of nutrients.

Besides growth, it is usually worth considering a death term for cells, in order to be able to describe the stationary phase of a cultivation. This is usually done by modifying the mass balance for X_V as follows:

$$\frac{dX_V}{dt} = \mu X_V - \mu_d X_V, \quad (6.18)$$

6.4 Bioreactor Operation Modes

where μ_d is the death rate of the cells and can be again expressed as a function of the extracellular environment. For example, Villiger et al. (2016a) modelled the death rate of CHO cells as a function of extracellular ammonia according to the following equation:

$$\mu_d = \frac{\mu_d^{max}}{1 + \left(\frac{k_{d,amm}}{c_{amm}} \right)^2}, \quad (6.19)$$

where μ_d^{max} is the maximum possible cell death rate and $k_{d,amm}$ a constant to relate the concentration of ammonia c_{amm} to the death rate. Lastly, antibody production can be expressed as

$$\frac{dc_{mAb}}{dt} = q_{mAb} X_V, \quad (6.20)$$

where q_{mAb} is the cell-specific productivity, which can either be assumed constant or can vary according to the environmental conditions. Usually, it is expressed as a function of the growth rate of the cells, since non-dividing cells usually show a higher productivity compared to rapidly proliferating cells, or process variables such as pH or osmolality.

Although the presented equations are generally valid for CHO cells, they can be modified to better represent the behaviour of other cell types (Tziampazis & Sambanis 1994). Moreover, as it will be discussed later, these equations can be complemented with additional ones to describe other aspects of bioprocessing such as the quality profiles of the secreted proteins. The following section describes how these equations can be incorporated in the mass balances that describe the different operation modes of the bioreactor.

6.4 BIOREACTOR OPERATION MODES

6.4.1 Batch Operation

The simplest operation mode for a stirred tank reactor – i.e., the batch mode – is based on the introduction of all reactants at the beginning of the operation, without any inlet or outlet stream (Figure 6.1) so that the general mass balance in Equation (6.2) reduces to

$$\frac{d(c_i V_R)}{dt} = r_i V_R, \quad (6.21)$$

which, by assuming constant reactor volume, V_R leads to

$$\frac{dc_i}{dt} = r_i, \quad (6.22)$$

with the initial conditions $c_i = c_i^0$ at time $t = 0$. In this special case of an ideal reactor system, the rate of accumulation of a species i in the reactor is equal to the rate of consumption or production of the same species.

We already indicated that the state-of-the-art operation of bioreactors used for the production of biopharmaceuticals is fed-batch. This operating mode can be modelled starting from the batch case (Section 6.4.1) by simply adding an inlet stream

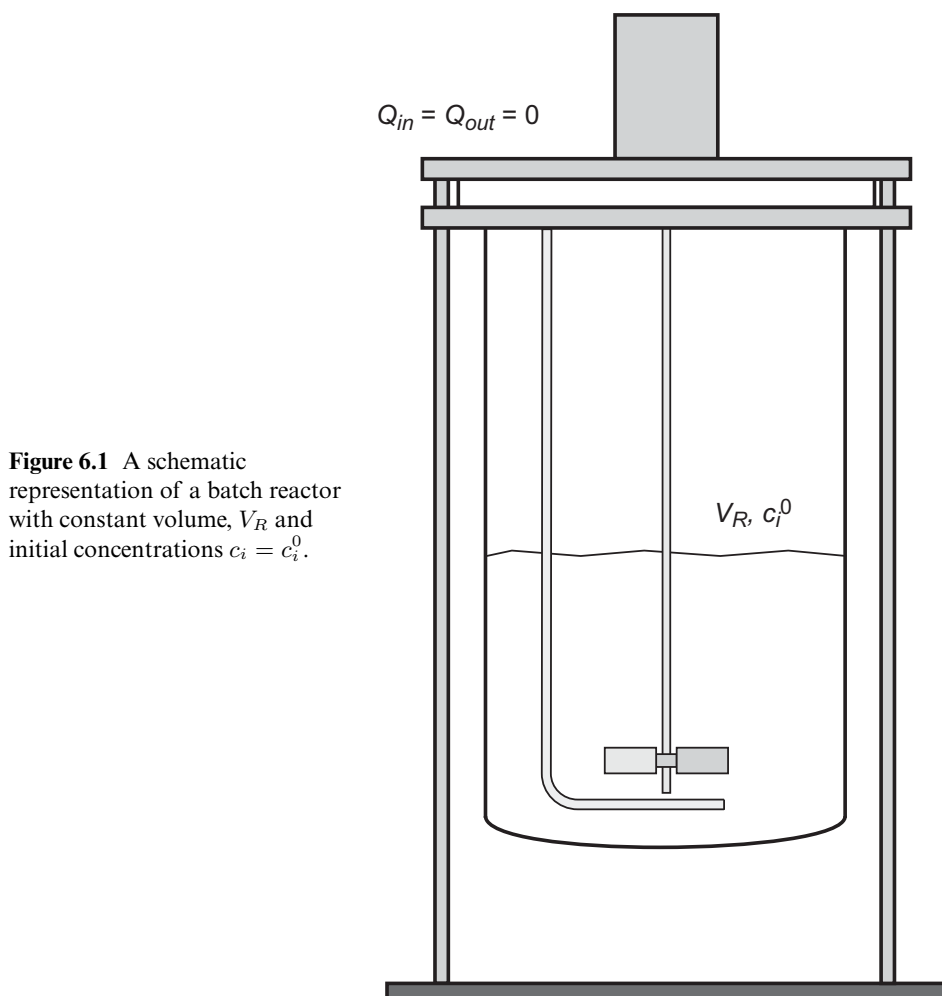


Figure 6.1 A schematic representation of a batch reactor with constant volume, V_R and initial concentrations $c_i = c_i^0$.

with volumetric flowrate, $Q_{in}(t)$ and composition defined by the molar concentrations, $c_{i,in}$ (Figure 6.2):

$$\frac{d(c_i V_R)}{dt} = Q_{in}(t)c_{i,in} + r_i V_R(t). \quad (6.23)$$

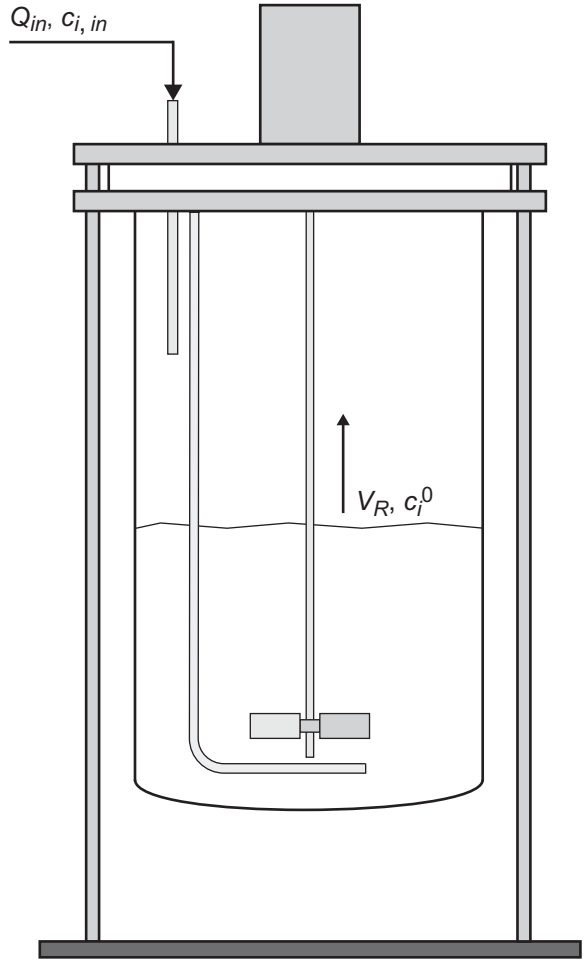
Since cells are not present in the inlet stream, the corresponding mass balance reduces to

$$\frac{d(V_R X_V)}{dt} = \mu X_V V_R - \mu X_V V_R. \quad (6.24)$$

For oxygen and carbon dioxide, the mass balance needs to take into account the volumetric mass transfer coefficient $k_L a [hr^{-1}]$, which, multiplied by the driving force, $\Delta C_{Gas} = (C_{Gas}^* - C_{Gas})$, gives the flowrate of the component coming in

6.4 Bioreactor Operation Modes

Figure 6.2 A schematic representation of a fed-batch reactor with the addition of a feed stream with volumetric flowrate, Q_{in} , and composition, $c_{(i,in)}$, with reactor volume, V_R , changing in time and initial concentrations $c_i = c_i^0$.



(out) the liquid phase, while the reaction term provides the flow of the same species consumed (produced) by the reactive processes inside the reactor:

$$\frac{dC_{gas}}{dt} = k_L a (C_{Gas}^* - C_{Gas}) + r_i V_R(t), \quad (6.25)$$

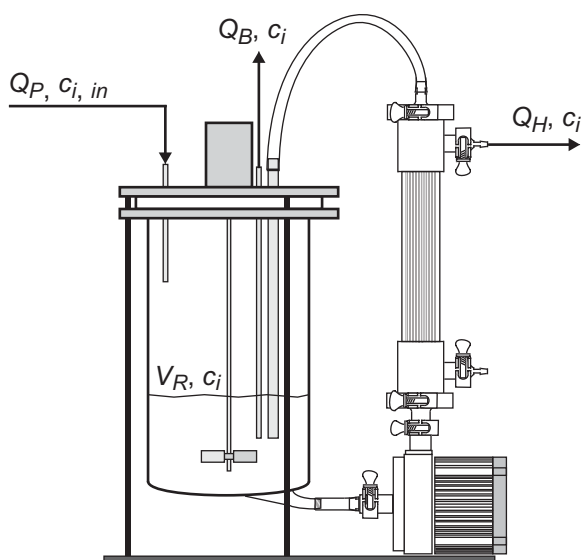
where C_{Gas}^* indicates the saturated concentration of the gaseous species of interest in the medium.

Due to the addition of the feed stream, the reactor volume changes during operation, as described by the following equation:

$$\frac{dV_R}{dt} = Q_{in}(t), \quad (6.26)$$

with the initial condition $V_R = V_R^0$ at $t = 0$.

Figure 6.3 A schematic representation of a perfusion bioreactor with the perfusion stream having volumetric flowrate, Q_P , and composition, $c_{i,in}$, and two outlet streams, the bleed and the harvest, with volumetric flowrates Q_B and Q_H , respectively.



It is worth mentioning that, in the case of smaller bioreactors, the amount of volume removed through sampling, Q_{out} , is not negligible and should be taken into account in the overall mass balance as follows:

$$\frac{dV_R}{dt} = Q_{in}(t) - Q_{out}(t). \quad (6.27)$$

Note that for fed-batch systems, neither $Q_{in}(t)$ nor $Q_{out}(t)$ are continuous functions with respect to time, since sampling and feeding are usually performed at discrete time points during the cultivation.

6.4.2 Continuous Perfusion Operation

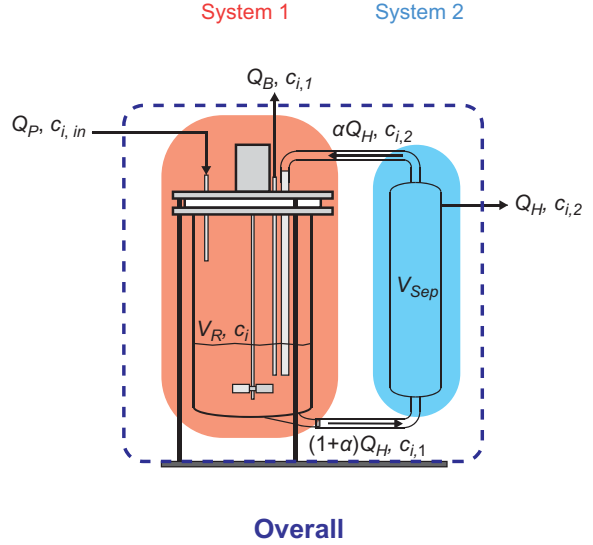
In Chapter 2, we have introduced the ideal CSTR model to simulate a stirred tank reactor operated in the perfusion mode, with the strong simplification that the cell retention device is placed inside the reactor tank itself. However, this does not usually hold true, since the filtration device is located outside the reactor and connected to it through proper piping. Accordingly, the cell retention device needs to be modelled as well. With reference to the perfusion bioreactor coupled to a tangential flow filtration (TFF) device shown in Figure 6.3, and assuming that the two outlet streams have the same composition as inside the reactor c_i , the overall mass balance and the mass balances for the individual metabolite species i can be written as follows:

$$Q_P = Q_B + Q_H \quad (6.28)$$

$$V_R \frac{dc_i}{dt} = Q_P c_{i,in} - (Q_B + Q_H) c_i + r_i V_R. \quad (6.29)$$

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Figure 6.4 Schematic representation of a perfusion bioreactor where the stirred tank (red) is modelled as an ideal CSTR system and the filtration device (light blue) can be modelled as a tubular ideal plug-flow reactor, PFR.



Assuming perfect cell retention (i.e., X_V in the harvest stream is equal to zero), the mass balance for the viable cell density, X_V reads

$$V_R \frac{dX_V}{dt} = -X_V Q_B + \mu X_V V_R - \mu_d X_V V_R. \quad (6.30)$$

In the metabolite mass balances (Equation (6.28)), we have assumed that the residence time in the external loop is much shorter than the time required for biological reactions to take place. Accordingly, the stream entering the loop is recycled unaltered back into the reactor, except for a portion corresponding to the harvest stream, $Q_H c_i$, which leaves the system. However, the residence time in the filtration module is, in some cases at the large scale, not negligible, and the same processes occurring inside the stirred tank are also taking place here and therefore need to be properly modelled. This can be done by considering the tank and the filtration module separately, as illustrated in Figure 6.4. The first one consists of a stirred vessel and is therefore reasonable to apply again the ideal CSTR model, as done earlier. For the second, however, we need to consider that the flow enters the module with a certain composition, $c_{i,1}$, equal to the corresponding composition c_i inside the stirred tank, which is then going to change as the flow proceeds along the hollow fibres and becomes, $c_{i,2}$ as the flow reaches the end of the filtration device and enters back into the stirred vessel. Since there is no stirring in this device, it appears reasonable to assume that the suspension progresses through the device like a liquid through a tube, which can be modelled using the ideal plug flow reactor (PFR) model (Fogler 2008, Froment et al. 1990). In a typical perfusion set-up, the three inlet and outlet stream flowrates – Q_P , Q_B and Q_H – are kept constant at the desired set-point value by the control system. The flow stream entering the filtration module is instead controlled by the pump of the TFF, and therefore, it is known and assumed equal to the harvest volumetric flowrate, Q_H , increased by a factor $(1+\alpha)$ in the following.

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The overall mass balance of the coupled systems is unchanged with respect to the previous case:

$$Q_P = Q_B + Q_H, \quad (6.31)$$

which is equivalent to what one would write with reference to the stirred tank alone (red in Figure 6.4):

$$Q_P + \alpha Q_H = Q_B + (1 + \alpha)Q_H. \quad (6.32)$$

About the single species i , the mass balances in Equation (6.28), are now changed and read as follows:

$$V_R \frac{dc_i}{dt} = Q_P c_{i,in} + \alpha Q_H c_{i,2} - (Q_B + (1 + \alpha)Q_H) c_{i,1} + r_i V_R, \quad (6.33)$$

while for the viable cell density, X_V , the mass balance for Equation (6.28) becomes

$$V_R \frac{dX_{V,1}}{dt} = X_{V,2} \alpha Q_H - X_{V,1} (Q_B + (1 + \alpha)Q_H) + \mu X_{V,1} V_R - \mu_d X_{V,1} V_R, \quad (6.34)$$

and for gaseous species like oxygen and carbon dioxide, we get

$$V_R \frac{dC_{Gas}}{dt} = k_L a (C_{Gas}^* - C_{Gas}) + \alpha Q_H c_{i,2} - (Q_B + (1 + \alpha)Q_H) C_{Gas} + r_i V_R. \quad (6.35)$$

Note that in agreement with the assumption of ideal mixing in the stirred tank, the conditions at the inlet of the filtration device are the same as those inside the reactor so that $c_{i,1} = c_i$ and $X_{V,1} = X_V$.

We now need to introduce the model of the filtration device in order to connect the outlet quantities, $c_{i,2}$ and $X_{V,2}$, to the inlet ones, $c_{i,1}$ and $X_{V,1}$ (light blue in Figure 6.4). As mentioned before, it is reasonable to model it as a tubular reactor with a distributed outlet liquid stream leaving the reactor while moving along its axis. This stream has the same composition as the liquid inside the loop, except for the viable cell density, which is equal to zero due to the filtering effect. In this frame, we obtain the following mass balances for the metabolite i :

$$\frac{d(Qc_{i,1})}{dt} = -c_{i,1} \frac{dQ}{dV} + r_i, \quad (6.36)$$

and for the viable cell density X_V :

$$\frac{d(QX_V)}{dV} = \mu X_{V,1} - \mu_d X_{V,1}, \quad (6.37)$$

where V is the axial coordinate along the filter going from $V = 0$ to $V = V_{sep}$.

Note that Equation (6.36) can be applied also to oxygen and carbon dioxide, since in this portion of the unit, there is no separate gaseous phase present, like for example, bubbles.

The flowrate of the liquid stream inside the filter decreases along the axial coordinate going from $(1 + \alpha)Q_H$ at the inlet to αQ_H at the outlet. Assuming as a first approximation that the harvest flux remains constant along the coordinate V , we obtain the following expression for the liquid flowrate inside the external loop:

$$Q = (1 + \alpha)Q_H - \alpha Q_H \left(\frac{V}{V_{sep}} \right) \quad (6.38)$$

6.5 Mechanistic Models and Their Implementation

Integrating Equations (6.36) and (6.37), from the initial condition $c_{i,1} = c_i$ and $X_{V,1} = X_V$ at $V = 0$, the outlet conditions $c_{i,2}$ and $X_{V,2}$ are obtained at $V = V_{sep}$.

In the case of lab-scale perfusion bioreactors, external flowrates in the range of 1.0–1.5 L/min are reported for both ATF and TFF systems, while the harvest rate typically ranges between 0.7 and 1.0 RV/day. For the case of a 2 L bioreactor system, this equals 1.4–2.0 L/day (0.07–0.0014 L/min), giving α values ranging between 700 and 1,100, which means that the rate through the external loop is about 1,000 times larger than the overall perfusion rate.

The average residence time, τ_{Sep} , in the cell retention device can be computed as follows:

$$\tau_{Sep} = \frac{V_{Sep}}{(1 + \alpha)Q_H}. \quad (6.39)$$

As discussed in the previous chapter, this is a critical parameter during scale-up since the environment in the cell retention device is not controlled in terms of pH, temperature, and composition, including oxygen concentration. The latter is likely the most critical one since the only oxygen available to the cell in this device is the one dissolved in the suspension at the entrance. As a rule of thumb, the residence time in the cell retention device has been reported for ATF devices to be critical if longer than about 60 seconds (Walther et al. 2019). Based on the model, it is possible to investigate the influence of the residence time on any of the cell culture parameters, including the dissolved oxygen concentration. This is a very useful tool that could provide flexibility in the scale-up. For example, if we are able to quantify the rate of oxygen consumption per passage in the cell retention device, we do not necessarily need to satisfy the rigid constraint of keeping the residence time constant during scale-up, at least with respect to oxygen consumption.

6.5 MECHANISTIC MODELS AND THEIR IMPLEMENTATION

6.5.1 Cell Culture and Cell Metabolism

Mechanistic or *white-box* models of biological systems, such as mammalian cell cultures, involve various degrees of structure and mathematical complexity, which tend to be higher compared to the reactor models introduced earlier (Sidoli et al. 2004). Generally speaking, modelling of biological organisms is aimed at improving the biological understanding of the complex cellular mechanisms, while also providing a framework for the control and optimisation of cell cultures. In the frame of bioreactor models, the main aim is to describe the interaction between cell-related parameters, such as cell-specific growth and productivity, to the operating conditions inside the bioreactor (Dhir et al. 2000, Dowd et al. 1999).

The first step of a mechanistic modelling approach involves translating the *a priori* knowledge into the mathematical description of the problem, which corresponds to the model development step, followed by the analysis of the model (Kiparissides et al. 2011). This contains an *a priori* identifiability analysis of all parameters or, in the case of dynamic, non-linear models, a sensitivity analysis that studies how much a change in the model output can be attributed to the variation of the different input parameters. This step is typically used to improve model accuracy and to identify the set of parameters that needs to be estimated from experimental data. The reliable estimation of parameters, however, is only feasible if a sufficiently rich and diverse

data set is available. Therefore, the use of optimal experimental design techniques is necessary to enable accurate parameter estimation and model validation (Kontoravdi et al. 2010). The last step requires the validation of the predictive capability of the model by comparison with a new set of independent experimental data.

In general, cell culture models necessitate a large number of differential and algebraic equations coupled with various parameters that need to be estimated. However, although we are still not in the position of developing models which couple a detailed description of the cell metabolism with that of the entire bioreactor, it is definitely possible to do so for more limited models aimed at the simulation of only specific quantities – for example, some structure characteristics of the target protein. In the following, we illustrate the power of mathematical mechanistic models, in particular, for the control and modulation of N-linked glycosylation patterns in mammalian cell culture processes (Jimenez del Val et al. 2011, Karst et al. 2017b, Villiger et al. 2016a).

6.5.2 N-Linked Glycosylation Modelling

N-linked glycosylation of therapeutic proteins is particularly well suited for the application of mathematical models. Contrary to other biological processes (e.g., metabolic and gene regulatory networks), the network of reactions and the enzymes involved in protein N-glycosylation are established, allowing the possibility for a relatively precise description of the entire system. On the other hand, the N-linked glycosylation patterns strongly affect the biological function of the protein, and therefore, their detailed understanding is crucial to optimise their therapeutic efficacy.

In eukaryotic cells, the first step of N-linked glycosylation occurs in the endoplasmic reticulum (ER). A large oligosaccharide (OS) structure ($\text{Glc}_3 \text{Man}_9 \text{GlcNAc}_2$) is transferred en bloc to a specific sequon (ASN-X-Ser) of the protein (Helenius & Aebi 2001). The site occupancy and the extent of additional modifications on the oligosaccharide chain depend on the efficiency of several subsequent enzymatic steps occurring in the secretory pathways. Factors that have an effect on the glycosylation process include the amount and localisation of various enzymes (kinases, transferases, permeases and epimerases), glycan–protein interactions (Losfeld et al. 2017), the pH in the lumina, and the flux and velocity of glycosylated proteins passing through the Golgi apparatus (Rivinoja et al. 2009, Roy 2009). Moreover, it is known that metal cofactors and the availability of nucleotide activated sugars (NS) affect the enzymatic kinetics of the oligosaccharide complex and many glycosyltransferases (Gramer et al. 2011, Kochanowski et al. 2008). Interference of these various enzymatic steps can result in differences in the glycosylation profile at a specific site (microheterogeneity). Several mathematical models have been reported, focusing mostly on glycan processing in the Golgi apparatus (Krambeck & Betenbaugh 2005, Umaña & Bailey 1997).

Recombinant monoclonal antibodies represent a special class of therapeutic proteins with respect to N-linked glycosylation. As they belong to the class of immunoglobulin G (IgG) proteins, they consistently bear N-linked glycosylation at the crystalline (Fc) region of both heavy chains with high occupancy, although they can also present other glycosylation sites in the variable region. The prevalent diantennary glycan structures are relatively simple compared to highly branched

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structures reported for various other glycoproteins. In the case of mAb Fc N-linked glycosylation, this allows us to consider only the microheterogeneity with a simplified network, which constitutes an ideal case for the development and application of a mathematical modelling framework in order to describe, predict and control N-linked glycosylation patterns. In a first proposal, this process cascade was described and simulated as a series of four ideally mixed compartments with the mass balances of the individual structures based on an enzyme-catalysed network (Umaña & Bailey 1997). Further expansion incorporated the addition of the fucosylation, galactosylation and sialylation steps ending up in a large network with thousands of substances and reactions (Krambeck & Betenbaugh 2005). Recent investigations were focused on dynamic models of monoclonal antibody glycosylation that link cellular metabolism via nucleotide sugar donors in order to couple the nucleotide sugar synthesis to the mechanistic glycosylation model (Jimenez del Val et al. 2011). This can be applied in a rather general way, and the dynamic model can be linked to any (of course, reasonable) unstructured or structured model, such as a simple cell culture model.

Mechanistic N-Linked Glycosylation Model In two exemplary case studies, Villiger et al. (2016b) and Karst et al. (2017b) used this model description and investigated the effect of manganese, galactose, culture pH and the ammonia concentration on the glycosylation pattern both in fed-batch and perfusion operations. In particular, they further developed the model of Jimenez del Val et al. (2011) and used it to describe the N-linked glycosylation trimming and elongation process based on the biosynthetic pathway within the Golgi shown in Figure 6.5. This model assumes that the first step of glycan processing taking place in the ER is fully completed – i.e., that every single mAb Fc region entering the Golgi has the same structure (M9). The Golgi apparatus itself is modelled as a single PFR, which includes the recycling of Golgi-resident proteins and transport equations for nucleotide sugars (Jimenez del Val et al. 2011). The mass balances of the different oligosaccharide structures OS_i ; nucleotides, N_k ; and nucleotide sugars, NS_k , are calculated considering the Golgi PFR model operating at steady state and the simplified glycosylation kinetic model shown in Figure 6.5, involving 43 reactions with 33 different glycan structures, as follows:

$$\frac{\delta [OS_i(z, t)]}{\delta t} = -\frac{4q}{\pi D_i^2} \frac{\delta [OS_i(z, t)]}{\delta z} - \sum_{j=1}^{NR} v_{i,j} r_j \quad (6.40)$$

$$\frac{\delta [N_k^{Golgi}(z, t)]}{\delta t} = -F_{T,k} - \frac{4q}{\pi D_i^2} \frac{\delta [N_k^{Golgi}(z, t)]}{\delta z} - \sum_{j=1}^{NR} v_{k,j} r_j \quad (6.41)$$

$$\frac{\delta [NS_k^{Golgi}(z, t)]}{\delta t} = -F_{T,k} - \frac{4q}{\pi D_i^2} \frac{\delta [NS_k^{Golgi}(z, t)]}{\delta z} - \sum_{j=1}^{NR} v_{k,j} r_j, \quad (6.42)$$

where D_i is the Golgi diameter, $v_{k,j}$ the stoichiometric coefficient, r_j the reaction rate, NR is the number of enzymatic reactions, q is the flowrate in the Golgi apparatus and

$F_{T,k}$ a term describing the flowrate of sugar precursors into the Golgi, which depends on the gradient of nucleotides and nucleotide sugars across the Golgi membrane:

$$F_{T,k}(z) = k_{T,k} [TP_k(z)] \left(\frac{[NS_k^{cylt}]}{k_{NS,k}^{cylt} + [NS_k^{cylt}]} \right) \left(\frac{[N_k^{cylt}]}{k_{N,k}^{cylt} + [N_k^{cylt}]} \right) \quad (6.43)$$

where $k_{T,k}$ represents the transport turnover rate and $[TP_k(z)]$ the concentration of the transport protein k at a Golgi position z . The transport proteins TP_k as well as the glycosyltransferases E_j are located along the Golgi with a Gaussian distribution:

$$[TP_k(z)] = [TP_k^{max}] \exp \left[-\frac{1}{2} \left(\frac{z - z_k^{max}}{\omega_k} \right)^2 \right] \quad (6.44)$$

$$[E_j(z)] = [E_j^{max}] \exp \left[-\frac{1}{2} \left(\frac{z - z_j^{max}}{\omega_j} \right)^2 \right], \quad (6.45)$$

with TP_k^{max} representing the peak concentration of a transport protein at a certain Golgi length z_k^{max} and E_j^{max} representing the peak concentration of a glycosyltransferase j at a certain Golgi position z_j^{max} .

As mentioned before, the 43 different reactions in Figure 6.5 are catalysed by seven enzymatic reactions whose kinetics is described in the following. First, $\alpha-1,2$ mannosidase I (ManI) and $\alpha-1,3/\alpha-1,6$ mannosidase II (ManII) remove mannose (MAN) units from the glycan structure in the earliest part of the pathway (Tabas & Kornfeld 1979, Velasco 1993). Their kinetics were assumed to follow the well-known Michaelis–Menten kinetics:

$$r_j = \frac{k_{f,j} [E_j] [OS_i]}{K_{d,i} \left(1 + \frac{[OS_i]}{K_{d,i}} + \sum_{z=1}^{NC} \frac{[OS_z]}{K_{d,z}} + \frac{[OS_{i-1}]}{K_{d,i-1}} \right)}, \quad (6.46)$$

with $k_{f,j}$ representing the turnover rate of the reaction and $K_{d,i}$ representing the dissociation constant of the specific donor-enzyme complex. Next, Villiger et al.

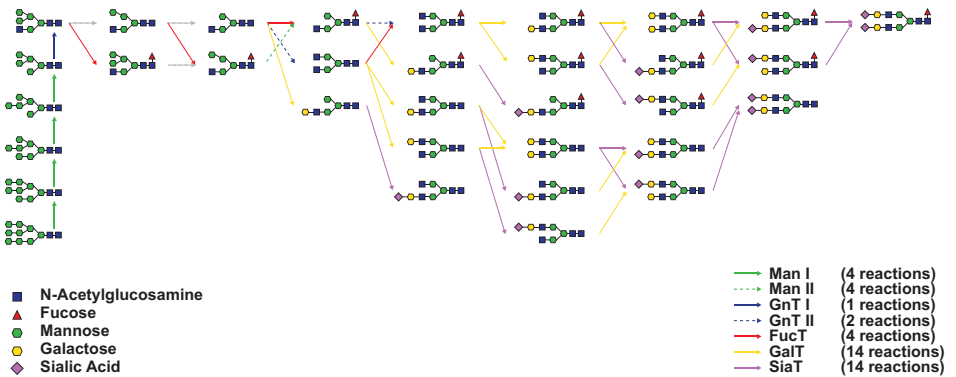


Figure 6.5 Simplified network describing mAb FC N-linked glycosylation used for the dynamic mechanistic models. The network consists of 33 different glycan structures (11 of them are measured), and 43 reactions catalysed by seven enzymes. © Reprinted from *Biotechnology Progress*, Thomas K. Villiger, Anaïs Roulet, Arnaud Périlleux, Matthieu Stettler, Hervé Broly, Massimo Morbidelli and Miroslav Soos, ‘Controlling the time evolution of mAb N-linked glycosylation: Part I: Microbioreactor experiments’, 2016, with permission from John Wiley and Sons

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(2016b) considered $\alpha - 1, 3$ N-acetylglucosaminyl transferase I (GnTI) and $\alpha - 1, 6$ N-acetylglucosaminyl transferase II (GnTII), which are responsible for the addition of N-Acetylglucosamine to the glycan chain (in total, three reactions) (Bendiak & Schachter 1987). Similarly, $\beta - 1, 4$ Galactosyltransferase (GalT) catalyses the addition of galactose to the glycan chain, and is involved in 14 such elongation reactions (Bendiak & Schachter 1987, McCracken et al. 2014). The enzyme kinetics of these three glycosyltransferases are assumed to be based on sequential-order bi-bi kinetics, and their activity is highly influenced by the presence of metal co-factors (Gramer et al. 2011). Of particular interest for the enzyme kinetics is the presence of manganese ions $[Mn^{2+}]$. The metal co-factor mediates the binding between the enzyme and the nucleotide sugar, which is necessary in order to provide the required conformational changes for the binding to the N-glycan substrate. After the formation of the modified N-glycan, the glycan structure is released first, followed by the manganese ion and a phosphate group that is released as a by-product (Lairson et al. 2008). This specific type of kinetics can be expressed as follows:

$$r_j = \frac{k_{f,j} [E_j] [Mn^{2+}] [NS_k^{Golg i}] [OS_i]}{K_{d,Mn^{2+}} + K_{d,k} K_{d,i} \left(1 + \frac{[Mn^{2+}]}{K_{d,Mn^{2+}}} + \frac{[Mn^{2+}]}{K_{d,Mn^{2+}}} \frac{[NS_k^{Golg i}]}{K_{d,k}} + \frac{[Mn^{2+}]}{K_{d,Mn^{2+}}} \frac{[NS_k^{Golg i}]}{K_{d,k}} \frac{[OS_i]}{K_{d,i}} \right. \\ \left. + \frac{[Mn^{2+}]}{K_{d,Mn^{2+}}} \frac{[NS_k^{Golg i}]}{K_{d,k}} \sum_z^{NC} \frac{[OS_z]}{K_{d,z}} + \frac{[OS_{i+1}]}{K_{d,i+1}} \frac{[N_k^{Golg i}]}{K_{d,Nk}} + \frac{[N_k^{Golg i}]}{K_{d,Nk}} \right)} \quad (6.47)$$

with $K_{d,k}$ representing the dissociation constant of the donor-enzyme complex, $K_{d,Mn^{2+}}$ representing the dissociation constant of the manganese-enzyme complex and $K_{d,Nk}$ representing the dissociation constant of the nucleotide-enzyme complex. Another glycosyltransferase involved in the reaction scheme is $\alpha - 1, 6$ fucosyltransferase (FucT) that catalyses the transfer of a fucose molecule to the glycan chain (overall, four reactions in the network). The last enzyme of this network is the $\alpha - 1, 6$ sialyltransferase (SialT), which is involved in the addition of sialic acid to the nucleotide chain. Overall, the enzyme is involved in 14 reactions of the network. The enzyme kinetics of both transferase enzymes, FucT and SialT, can be approached by applying a random-order bi-bi kinetic rate, expressed as follows:

$$r_j = \frac{k_{f,j} [E_j] [NS_k^{Golg i}] [OS_i]}{K_{d,k} K_{d,i} \left(1 + \frac{[NS_k^{Golg i}]}{K_{d,k}} + \frac{[OS_i]}{K_{d,i}} + \sum_{z=1}^{NC} \frac{[OS_z]}{K_{d,z}} + \frac{[NS_k^{Golg i}]}{K_{d,k}} \frac{[OS_i]}{K_{d,i}} \right. \\ \left. + \frac{[NS_k^{Golg i}]}{K_{d,k}} \sum_{z=1}^{NC} \frac{[OS_z]}{K_{d,z}} + \frac{[N_k^{Golg i}]}{K_{d,Nk}} \frac{[OS_{i+1}]}{K_{d,i+1}} + \frac{[N_k^{Golg i}]}{K_{d,Nk}} + \frac{[OS_{i+1}]}{K_{d,i+1}} \right)} \quad (6.48)$$

Having summarised the enzyme kinetics for the N-linked glycosylation network, the next step is to evaluate the activity of the glycosylation enzymes as a function of the environmental conditions prevailing in the Golgi apparatus. Typically, the Golgi pH (pH_{Golgi}) plays the most important role with respect to the enzymes activity (Rivinoja et al. 2009). In mammalian cells, the Golgi pH, referred to as pH_{Golgi} , is slightly acidic (5.9–6.5) and well regulated in normal conditions. It is well known that extracellular ammonia and other bases that easily diffuse through the cell membranes

can have a significant effect on the intracellular pH_{Golgi} , resulting in mislocalisation and reduced activity of the glycosylation enzymes. However, in most cases, it is assumed that the enzyme position across the Golgi apparatus remains unaffected, and only the enzyme activity is changed by varying intracellular pH. With respect to mammalian cell cultures, the effect of increasing ammonia concentration in the cell culture leading to larger values of the intracellular pH_{Golgi} can be expressed by the Henderson–Hasselbalch equation. The pH dependency of the activity of Golgi resident enzymes was estimated by Villiger et al. (2016c) from literature data of in vitro enzymatic assays, determining an optimal pH for the activity of Golgi-resident enzymes pH_{opt}^{Golgi} of 6.59. The turnover rate constant as well as the Henderson–Hasselbalch equation for the pH_{Golgi} can therefore be expressed as follows:

$$k_{f,j} = k_{f,j}^{max} \exp \left[-\frac{1}{2} \left(\frac{pH^{Golgi} - pH_{opt}^{Golgi}}{\omega_{f,j}} \right)^2 \right] \quad (6.49)$$

$$pH^{Golgi} = pK_A^{Golgi} + \log \left(\frac{[AMM]}{N_A^{Golgi} - [AMM]} \right), \quad (6.50)$$

with $k_{f,j}^{max}$ representing the maximum turnover rate of a specific reaction, $\omega_{f,j}$, the width of the pH activity profile of E_j , pK_A^{Golgi} the apparent pK_A value of the Golgi, $[AMM]$ the concentration of ammonia and N_A^{Golgi} the ammonia-associated Golgi pH constant. Due to the strong connection of the pH^{Golgi} and the pK_A^{Golgi} , these parameters cannot be estimated *a priori*, and therefore, Villiger et al. (2016b) provided a reasonable buffer capacity by setting the pK_A^{Golgi} equal to 7.5 at a pH value of 6.5 and used only the N_A^{Golgi} as a fitting parameter.

This detailed dynamic glycosylation model needs to be linked to a cell culture model that describes the external measured parameters, such as viable cell density, cell-specific productivity, ammonia production and intracellular nucleotide sugars and, thus, links these parameters to the one involved in the dynamic glycosylation model. Villiger et al. (2016b) and Karst et al. (2017b) proposed to use simple unstructured cell culture models to describe both N-linked glycosylation within fed-batch and perfusion bioreactor operations to overcome practical parameter identifiability and error propagation to the glycosylation model. Therefore, in both applications, the mass balances were kept simple and with the least amount of parameters to describe different operating conditions. In the following section, we focus on the connection of the dynamic N-linked glycosylation model to the different kind of operating conditions.

Fed-Batch Culture for IgG Production Villiger et al. (2016b) connected the dynamic model to describe N-linked glycosylation with a simple unstructured cell culture model used to complete the bioreactor model. The mass balances describing the culture volume and concentrations of manganese and galactose in the extracellular medium take into account perfect mixing and negligible consumption:

$$\frac{dV_R}{dt} = Q_{in} - Q_{out} \quad (6.51)$$

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$$\frac{dV_R [Mn^{2+}]}{dt} = Q_{in} [Mn_{in}^{2+}] - Q_{out} [Mn^{2+}] \quad (6.52)$$

$$\frac{dV_R [Gal]}{dt} = Q_{in} [Gal_{in}] - Q_{out} [Gal], \quad (6.53)$$

where Q_{in} is the feed volumetric flowrate, Q_{out} the flowrate out of the reactor (sampling), $[Mn_{in}^{2+}]$ and $[Gal_{in}]$ the inlet concentration of manganese and galactose, and $[Mn^{2+}]$ and $[Gal]$ the corresponding concentrations inside the reactor.

In addition, the model includes differential equations for the viable cell density X_V , the dead cell density X_d , and lysed cell population X_l as a function of the growth rate μ , death rate μ_d and lysis rate μ_l :

$$\frac{dV_R X_V}{dt} = \mu V_R X_V - \mu_d V X_V - Q_{out} X_V \quad (6.54)$$

$$\frac{dV_R X_d}{dt} = \mu_d V_R X_d - \mu_l V X_d - Q_{out} X_d \quad (6.55)$$

$$\frac{dV_R X_l}{dt} = \mu_l V_R X_d - Q_{out} X_l. \quad (6.56)$$

The growth and dead rates can be assumed to be functions of the ammonia concentration in the extracellular medium and can be described as follows:

$$\mu = \mu_{max} \left(\frac{K_{\mu, AMM}}{K_{\mu, AMM} + [AMM]} \right), \quad \text{for } t < 7 \text{ days} \quad (6.57)$$

$$\mu_d = \frac{\mu_d^{max}}{1 + \left(\frac{K_{d, AMM}}{[AMM]} \right)^2}, \quad (6.58)$$

with μ_{max} and μ_d^{max} the maximum growth and death rate, respectively; $K_{\mu, AMM}$ the ammonia growth inhibition constant, and $K_{d, AMM}$ the ammonia death inducing constant.

The lysis rate μ_l was empirically introduced as a fitting parameter to provide a mean to remove dead cells from the system in order to correctly describe the behaviour of the cell viability during cultivation.

In a next step, the ammonia concentration can be described as a function of the cellular-growth-rate, and maintenance-dependent parameters (Jang & Barford 2000) as follows:

$$\frac{dV_R [AMM]}{dt} = q_{AMM} V_R X_V - Q_{out} [AMM], \quad (6.59)$$

with q_{AMM} to be

$$q_{AMM} = \frac{\mu}{Y_{\mu, AMM}} + m_{AMM}, \quad (6.60)$$

where $Y_{\mu, AMM}$ represents a growth-dependent ammonia yield coefficient, and m_{AMM} represents a maintenance-related coefficient.

In addition, the model includes the mass balance for the target mAb. In the case of Villiger et al. (2016b), the specific productivity q_{mAb} of the employed cell line was observed to be constant with respect to the integral viable cell densities, but it followed a bell-shaped curve as a function of the extracellular pH

(Ozturk & Palsson 1991). The specific productivity can be consequently described with a Gaussian distribution with optimal culture pH at about 7.15, as experimentally determined:

$$\frac{dV_R C_{mAb}}{dt} = q_{mAb} V_R X_V - Q_{out} C_{mAb} \quad (6.61)$$

$$q_{mAb} = q_{mAb}^{max} \exp \left[-\frac{1}{2} \left(\frac{pH - pH_{opt}}{\omega_{q,mAb}} \right)^2 \right], \quad (6.62)$$

where $\omega_{q,mAb}$ describes the width of the specific productivity as a function of pH.

Moreover, the cell culture model has to include the mass balances of the nucleotide sugars. There are two main classes of nucleotide sugars. A first group that is typically produced in the cytosol – such as uridine diphosphate N-acetylglucosamine (UDP-GlcNAc), guanosine diphosphate fucose (GDP-Fuc) and uridine diphosphate galactose (UDP-Gal) – and the second that is synthesised in the cell nucleus that includes cytidine diphosphate N-acetylneuraminic acid (CMP-Neu5Ac). After synthesis, the nucleotide sugar precursors are transferred to the ER and to the Golgi apparatus via so-called active anti-transporters. The mass balances for the intracellular metabolites can be expressed in a simplified form with only one growth-dependent and one maintenance-dependent parameter. The general mass balance for nucleotide sugars can be stated as follows:

$$\frac{d[N S_k^{cyl}]}{dt} = \frac{\mu}{Y_{NS,k}} + m_{NS,k}, \quad (6.63)$$

with a growth-dependent yield coefficient $Y_{NS,k}$ and a maintenance-dependent parameter $m_{NS,k}$. In the case, where a precursor is added within the medium feed, such as galactose, the incorporation rate of that nucleotide sugar, $v_{NS,k}$ is assumed to be proportional to the concentration in the cytosol and the external concentration of the monosaccharide in the medium, referred to as MS_k . Consequently, the extended mass balance can be expressed as follows:

$$\frac{d[N S_k^{cyl}]}{dt} = \frac{\mu}{Y_{NS,k}} + m_{NS,k} + k_{NS,k} \left(\frac{[MS_k]}{K_{NS,k}^{MS}} - [N S_k^{cyl}] \right), \quad (6.64)$$

where $K_{NS,k}^{MS}$ represents the equilibrium constant describing the equilibrium between the monosaccharide concentration in the medium and in the cytosol.

After collecting data from various micro-bioreactor system runs (including feeding time points, online pH, DO, and sparging rates; analytical results of N-linked glycosylation; and intracellular components), Villiger et al. (2016b) used cell- and process-specific values as initial and boundary conditions for the structured glycosylation model. In addition, the set of unstructured ordinary differential equations of the cell culture model was solved and used as a link for the initiation of the glycosylation model, which resulted in a system of partial differential equations (PDEs). In total, the experimental data of 12 experiments with a total of 48 different glycan structures were used to tune the parameter space of the glycosylation model. Villiger et al. (2016b) tested the predictive capability of the model framework for different feeding strategies, resulting in a vast modulation of the glycosylation patterns, and the model results were able to predict the main glycan structures in both experiments, as shown in the top part of Figure 6.6. Comparing the experimentally measured glycan forms and the model predictions revealed that the described

Model predictions

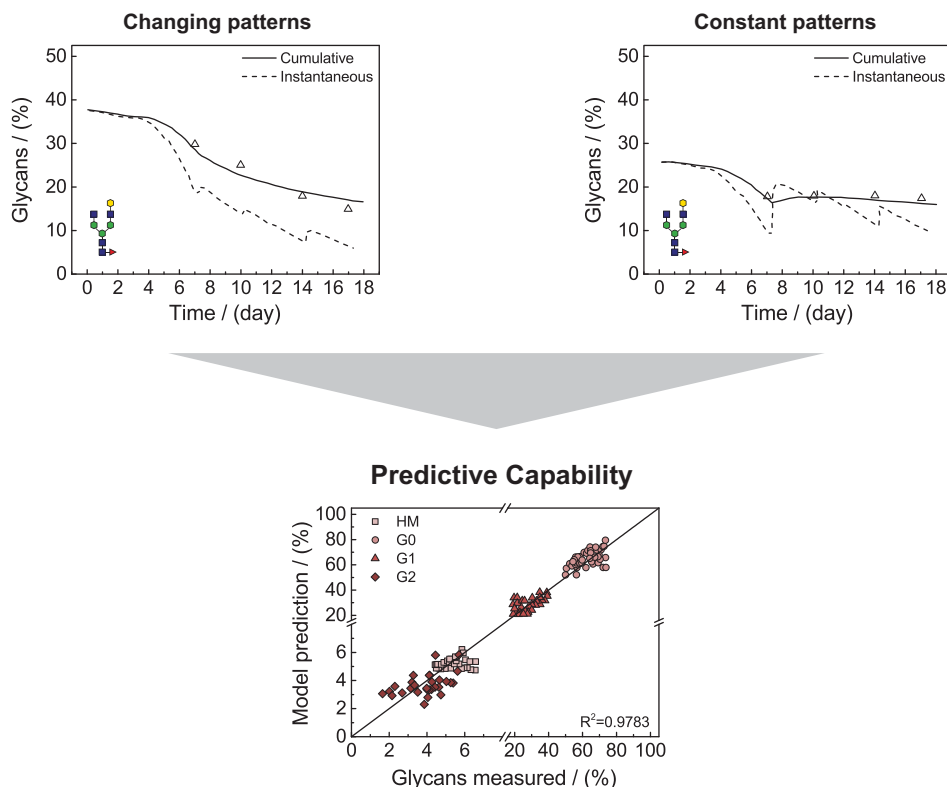


Figure 6.6 Evaluation of model performance. Predicted time evolution of an exemplary glycan structure (G1) for two different culture conditions (upper left and upper right), resulting in either a changing or constant glycan pattern as a function of time. The dotted line represents the model prediction for the instantaneous produced glycans, while the solid line represents the cumulative produced glycans. Below, the overall predicted versus measured mAb Fc glycosylation is presented for the most abundant glycan species of the overall data set. © Reprinted from *Biotechnology Progress*, Thomas K. Villiger, Ernesto Scibona, Matthieu Stettler, Hervé Broly, Massimo Morbidelli and Miroslav Soos, 'Controlling the time evolution of mAb N-linked glycosylation: Part II: Model-based predictions', 2016, with permission from John Wiley and Sons

modelling framework was able to predict glycosylation processes for a given process platform (bottom part of Figure 6.6). For fed-batch systems, the N-linked glycosylation profiles of the antibodies change with respect to time due to the accumulation of waste metabolites like ammonia (top left part of Figure 6.6). The model was able to define a feeding strategy that would result in constant glycosylation patterns in time through the addition of galactose and manganese at the latest stages of the culture (top right part of Figure 6.6), indicating that mechanistic models can be used as a robust tool for the optimisation towards defined glycosylation patterns for a given fed-batch process (Figure 6.6). A drawback of this deterministic unstructured cell culture model, which represents the basis of the mechanistic glycan model, is the very limited space of its predictive capability and can be only considered as valid for a stable platform process.

Perfusion Culture for IgG Production Karst et al. (2017b) expanded the application of the dynamic mechanistic N-linked glycosylation model to perfusion cultures. Therefore, a DoE with varying cell density (20×10^6 cells/mL (–), 40×10^6 cells/mL (0), and 60×10^6 cells/mL (+)), and supplementation of manganese ($0.01 \mu\text{M}$ (–), $0.1 \mu\text{M}$ (0), and $1.0 \mu\text{M}$ (+)), as well as galactose (0 mM (–), 5 mM (0), and 10 mM (+)) was constructed. Increasing the viable cell density represents a valuable tool to vary the ammonia concentration within the culture broth, which has been shown to be one of the most effective inhibitors of glycosylation, while manganese and galactose have been reported to support the production of more complex glycans, such as the galactosylation and sialylation of the terminal glycan units. While Mn^{2+} represents an essential primary co-factor for GnTI, GnTII and GalT, galactose is the sugar precursor of the UDP-Gal nucleotide sugar donor (Gramer et al. 2011, Ramakrishnan et al. 2004). Both high concentration of nucleotide sugar donors and high co-factor concentration are expected to enhance the dynamics within the glycosylation network. Due to the different reactor and process dynamics in perfusion, Karst et al. (2017b) used a modified version of the unstructured cell culture model. The overall reactor balance is the same as in Equation (6.51), but in contrast to the fed-batch reactor, Q_{out} represents the sum of cell-removing bleed and cell-free harvest flowrates

$$Q_{out} = Q_{Harvest} + Q_{Bleed}, \quad (6.65)$$

which results in the overall mass balance

$$\frac{dV_R}{dt} = Q_{in} - Q_{Harvest} - Q_{Bleed}. \quad (6.66)$$

Due to continuous media exchange and removal of cells in the bleed stream, the operation of perfusion bioreactors allows to maintain high cell viability close to 100 per cent; therefore, cell death and lysis can be neglected in the overall mass balance of the viable cell density:

$$\frac{d(V_R X_V)}{dt} = \mu V_R X_V - Q_{Bleed} X_V, \quad (6.67)$$

with μ the growth rate (day^{-1}) and V_R the reactor volume (L), and X_V the viable cell concentration (10^6 cells/mL). At steady state in a perfusion run, the viable cell density is kept constant, and therefore, the volumetric bleed rate can be expressed as follows:

$$Q_{Bleed} = \mu V_R. \quad (6.68)$$

For the growth rate μ – as well as the mass balances of galactose, manganese, ammonia – and the mAb concentration, Equations (6.52), (6.53), (6.57), (6.59) and (6.61) are still valid but have to be adjusted to the corresponding Q_{out} . With respect to the concentration of the nucleotide sugar donor UDP-Gal, Karst et al. (2017b) used a simplified version of Equation (6.64), which was sufficient to reproduce the experimental data without the addition of more parameters:

$$\begin{aligned} \frac{d[UDP - Gal_{Gal}^{cyt}]}{dt} = & m_{UDP-Gal, Gal} \\ & + k_{UDP-Gal, Gal} \left(\frac{[MS_{Gal}]}{K_{UDP-Gal, Gal}^{Gal}} - [UDP - Gal_{Gal}^{cyt}] \right). \end{aligned} \quad (6.69)$$

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As in the case of the fed-batch model, the parameters derived from the unstructured perfusion model provide the initial and boundary conditions for the structured N-linked glycosylation model. Nucleotide and nucleotide sugar concentrations (except for UDP-Gal) were assumed to be constant during the investigated conditions since continuous systems are known to provide a steady-state condition both intra cellularly and extracellularly. As illustrated in the section about mechanistic N-linked glycosylation model, the kinetics of the enzymatic reactions – in particular, Equations (6.20) and (6.23) were linked to the concentrations of Mn^{2+} and AMM , as well as to the availability of the sugar precursors. This enabled the simulation of the effect of the different operating conditions on the N-linked glycosylation patterns. Model-specific constants and other Golgi-specific parameters were taken from the literature (Jimenez del Val et al. 2011, Villiger et al. 2016b).

With respect to the experimental investigation, a DoE approach was combined with the sequential screening of several set points within long-term perfusion cultures. The perfusion runs were performed at a fixed harvest rate of 1.0 RV/day, and VCD set point changes were enabled by varying the fraction of base to enriched media between 100/0 per cent (20×10^6 cells/mL), 65/35 per cent (40×10^6 cells/mL) and 30/70 per cent (60×10^6 cells/mL), respectively, to account for higher metabolite consumption. The DoE was based on a two-level full factorial design, which was carried out in sequential perfusion culture experiments. In Table 6.1, the experimental conditions of the full factorial design with varying VCD, galactose and manganese concentration as well as the resulting patterns of the main glycoforms (such as HM, G0, G1 and G2) are summarised.

The experimental results revealed a high variability of glycoforms as a function of the operating conditions as shown in Table 6.1. While high-mannose (HM) structures were, in particular, found at higher VCDs due to the higher ammonia concentrations, ungalactosylated and galactosylated structures were widely spread as a result of the different operating conditions. With respect to the DoE conditions, for G0, a modulation space was found between 36 per cent (minimum) and 70 per cent (maximum), while G1 ranged from 18 to 48 per cent, and G2 from 1 per cent to 11 per cent. The systematic modulation approach was then, in a next step, supported by the training of the mechanistic glycosylation model, based on the first eight DoE experiments. The mechanistic model showed good agreement with the experimental measured glycan structures – in particular, for highly modulated structures, such as G0 and G1. The model was capable of reproducing both the expected inhibitory effect of ammonia/high VCD and the enhancing effect of manganese and galactose, as shown in Figure 6.7. Six cell-specific parameters of the unstructured cell culture model were fitted: the maximum cellular growth rate, μ_{max} ; the ammonia growth inhibition constant, $K_{\mu, AMM}$; the ammonia and antibody specific productivity, q_{AMM} and q_{mAb} ; and the maintenance coefficient of UDP-Gal $m_{UDP-Gal}$; as well as the the equilibrium constant of the galactose UDP-Gal equilibrium $K_{UDP-Gal, Gal}^{Gal}$.

On the other hand, the mechanistic glycosylation model required the estimation of seven parameters to correctly reproduce the glycosylation profiles within the entire DoE: the dissociation constants for Golgi resident enzymes and the term N_A^{Golgi} in Equation (6.50) that represents the transport of ammonia to the Golgi apparatus.

In a next step, the model was used to predict an independent set of experimental conditions referred to as Centre, Outlier and Edge in Table 6.1. The trained model was, indeed, able to predict the resulting glycosylation patterns and, therefore, showed the capability for both intrapolation and extrapolation from the operating

Table 6.1 Experimental conditions of the full-factorial DoE as well as a middle point, an outlier and an edge point of the DoE with varying VCD, Gal and Mn^{2+} concentrations performed in sequential long-term perfusion cultures and the resulting steady-state glycan distributions in terms of the main glycans – such as HM, G0, G1 and G2. © Reprinted from *Biotechnology and Bioengineering*, Daniel J. Karst, Ernesto Scibona, Elisa Serra, Jean-Marc Bielser, Jonathan Souquet, Matthieu Stettler, Hervé Broly, Miroslav Soos, Massimo Morbidelli and Thomas K. Villiger, ‘Modulation and modeling of monoclonal antibody N-linked glycosylation in mammalian cell perfusion reactors’, 2017, with permission from John Wiley and Sons.

DoE conditions	VCD / (10^6 cells/mL)	Gal / (mmol/L)	Mn^{+2} / ($\mu\text{mol/L}$)	HM / (%)	G0 / (%)	G1 / (%)	G2 / (%)
– – –	20	0	0.01	1.2	61.1	31.7	3
– + –	20	10	0.01	1.1	49	40.9	5.6
– – +	20	0	1	1.4	40.6	45.4	8.6
– + +	20	10	1	1.6	35.7	47.7	10.9
+ – –	60	0	0.01	6.2	70.6	18.1	1.1
+ + –	60	10	0.01	3.7	64.5	26.3	2.2
+ – +	60	0	1	2.8	47.6	39.4	6.1
+ + +	60	10	1	3.1	43.8	41.2	6.8
Center	40	5	0.1	1.9	38.3	46.2	9
Outlier	40	20	2	1.1	31.3	50.6	12.7
Edge	40	0	0.01	2.8	62.8	28.2	1.1

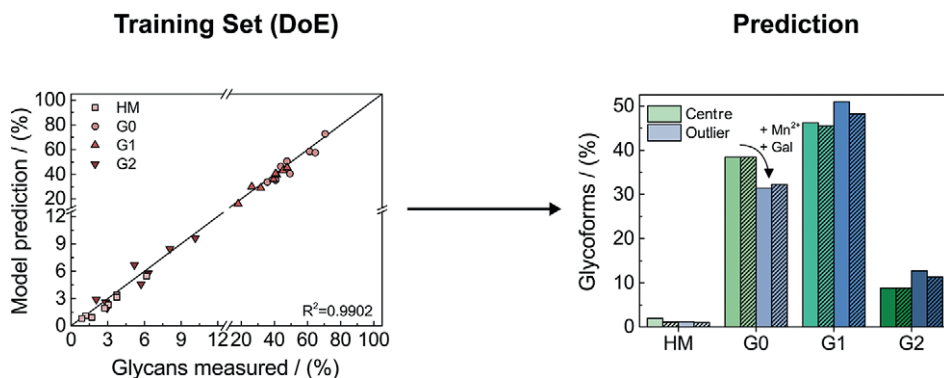


Figure 6.7 Evaluation of model performance in a perfusion application. On the left, the overall predicted versus measured mAb Fc glycosylation for the training set (two-level full-factorial DoE) is shown for the most abundant glycan species of the overall data set of Karst et al. (2017b). The trained model was used in a second step to predict the glycan patterns for another set of operating conditions. On the right, the experimental (plain) and predicted (stripped) glycan distribution at the centre (green) and the outlier point (point) in Table 6.1 are compared.

conditions covered by the DoE. For illustrative purposes, the experimental and predicted glycosylation patterns for the Centre and the Outlier operating conditions are compared in Figure 6.7.

From a more general point of view, these studies confirm the strength of mechanistic models with respect to process development, also in the case of biological systems, and they underline the power of the combination of unstructured and structured models. Glycan processing is a precisely described sequence of enzymatic steps that shows a high sensitivity with respect to environmental conditions. By combining this model with an unstructured cell culture model that is often kept rather simple, it is possible to train a mechanistic model that predicts N-linked glycosylation for both fed-batch and perfusion processes. Of course, the model predictions are only valid for suitable operating ranges of the given fed-batch or perfusion platform process. Nevertheless, this fact is important with respect to perfusion process development, which was discussed in Chapter 4. These examples underline the possibility of fine-tuning product quality of a given platform process with proper feed adjustments.

6.6 STATISTICAL AND HYBRID MODELLING

In this section, we discuss the application of statistical and hybrid modelling approaches to cell culture processes. The latter approach combines mechanistic or knowledge-driven models, as described in the previous section, with statistical or data-driven techniques. While mechanistic modelling approaches are derived from physical, chemical or biological principles, statistical approaches, often referred to as *black-box* or *empirical* models, search for data-driven correlations between input and output variables without involving any consideration based on first principles. More advanced approaches of this type, including time series analysis and historical prediction models, have also been developed to describe process dynamics.

Data-driven approaches have been proven to accurately express process trajectories in many areas of engineering, including bioprocessing (Bailey & Ollis 1986, Mercier et al. 2014). The data typically available in bioengineering and biotechnology

feature all of the four Vs of big data characteristics (Narayanan et al. 2019b) – namely volume, due to high-throughput and spectroscopic data; velocity, due to online measurements; veracity, as a result of the significant noise due to the living process nature and to the limited control, particularly at the small experimental scale; and variety, as a result of the heterogeneity of processes and products as well as devices, apparatuses and data management systems used along the lifecycle of process development. Moreover, the relationships between biological variables often are not completely understood, are not thoroughly measured and do not follow simple rate laws. Therefore, statistical methods provide an attractive way to efficiently extract information from such complex data and are a valuable tool to provide first insights into the bioprocess, highlight important relationships and influential variables, and generate hypotheses for subsequent modelling approaches (Sokolov et al. 2018). Statistical modelling offers a versatile toolbox ranging from simple multiple regression for uncorrelated regressors, multivariate analysis accounting for correlation amongst the regressors and non-linear machine learning approaches.

Despite the limited structural relation to the biotechnological process, the successful generation of these models depends on a clear understanding of the underlying model definitions and assumptions as well as the possibilities to tailor these to the nature of bioprocess data.

Statistical models are often used to support experimental design as well as to analyse the corresponding experimental data. The so-called design of experiments (DoE) approach facilitates the identification of relationships between given input and output variables, while placing experimental conditions as suitable combinations of several input variables at different levels. Such a DoE approach is significantly more efficient with regards to the experimental effort compared to classical approaches testing one variable at a time (Sommeregger et al. 2017). Moreover, combined variable testing is more likely to find important interaction or limitation effects amongst tested variables. This approach can also be utilised in a sequential way to iteratively identify relevant process variables and eventually define the so-called process design space in which these shall be controlled to maintain the targeted product quality in a defined specification range (Politis et al. 2017, Steeno 2010, Von Stosch & Willis 2017). In general, such quantitative approaches supporting the experimental effort during process development as well as providing new routes for knowledge generation are highly supported by the PAT and QbD initiatives of the regulatory authorities (Mercier et al. 2016, Rathore et al. 2016). However, solely statistically driven DoEs can be extremely laborious due to the amount of information which has to be tested (input variables and their ranges) and the limitations of the underlying linear models to withdraw relevant information on non-linear dynamic processes. Therefore, prior knowledge on central variables, their ranges and their interrelationships as well as advanced techniques to account for the non-linear and multivariate process nature are highly relevant to efficiently use experimental resources and already-available information. Hence, several applications and improvements of the classical DoE approaches have been presented in the literature in order to decrease the experimental effort and build more reliable and more generically applicable model solutions (Brühlmann et al. 2017b, Chakrabarty et al. 2013, Von Stosch & Willis 2017).

Besides the classical *white-box* (mechanistic) and *black-box* (statistic) modelling approaches, so-called *grey-box* or *hybrid* models present a compromise between mechanistic and statistic strategies by combining the available knowledge (such as simple unstructured cell culture models) with data-oriented modelling techniques

6.6 Statistical and Hybrid Modelling

and have been shown to often provide better predictions than pure black-box or pure white-box approaches (Narayanan et al. 2019a).

6.6.1 Multivariate Data Analysis Tools

In the case of large data sets with various independent variables, multivariate data analysis (MVDA) represents a useful tool to handle the high dimensionality and correlation between the different variables. Multivariate data analysis basically includes a set of tools that deal with more than one independent variable at a time and aim to perceive the correlations between such variables. In particular, principal component analysis and partial least square regression analysis extract important information, simplify datasets and estimate relationships between process variables (Ketaneh et al. 2005, Mercier et al. 2014). These techniques have been applied in investigations such as peptone screening batch fault detection, cell culture media screening and scale-down model qualification (Brunner et al. 2018, Gunther et al. 2007, Luo & Chen 2007, Ryan et al. 2010, Sokolov et al. 2017c, Tsang et al. 2014), and are discussed in the next sections.

Principal Component Analysis Extensive data sets, including the measurements of many variables, are increasingly common in cell culture processes due to the availability of more advanced process monitoring tools including omics approaches, high-throughput systems and online monitoring. Principal component analysis (PCA) is a common tool to reduce the number of variables into a smaller set of orthogonal principal components (PCs) while retaining most of the information of the original data set and separating it from process noise. PCA generally can be used as a visualisation tool to help identify groups, trends and central interrelationships (Jolliffe 2005, Meglen 1992). We can summarise the four goals of PCA as follows:

1. Extraction of the most important information from the data
2. Simplification of the visualisation and description of the data set benefiting from the highly correlated process nature
3. Analysis of the underlying correlation structure of the observations and the variables
4. Compression of the data dimensions and noise level by keeping only the important information

In particular, PCA decomposes the data matrix X with M variables and N observations. The data set can be represented either as M N -dimensional vectors $x_1, \dots, x_M$ or in a $(N \times M)$ matrix, X .

The goal of PCA is to reduce the dimension from M to A , with A typically much smaller than M , while retaining maximal information content. This can be done by finding linear combinations of the original variable x_m ($m = 1, 2, \dots, M$) with the coefficients $p_{m,a}$ ($a = 1, 2, \dots, A$), called principal components (PCs). These PCs are orthogonal to one another and are derived such that they explain maximal variance of the given data sets. By solving this maximisation problem, it can be found that p_a ($a = 1, 2, \dots, A$) are the eigenvectors of the covariance matrix, S , of the data corresponding to the A largest eigenvalues λ (Jolliffe 2005). The elements of the eigenvectors p_a are commonly called *loadings*, and the elements of the linear combinations Xp_a ($= t_a$) are called the *scores*. From a graphical perspective, an eigenvector represents the identified direction along which the maximal fraction of

the variance, not explained by the previously defined eigenvectors, can be explained, which is indicated by the corresponding eigenvalue. In particular, the first and the second principal component represent the eigenvectors having the highest eigenvalues, and consequently contain the most information – e.g., the axis along which the data is most spread out.

For PCA, the components can be obtained by performing a single-value decomposition (SVD) of the data matrix X as follows (Bartholomew 2010):

$$X = U\Delta V^T + E, \quad (6.70)$$

with U representing the $(N \times A)$ matrix of the left singular vectors and V representing the $(M \times A)$ matrix of the right singular vectors, often referred to as loading matrix (columns of V represent the principal directions) and Δ representing the diagonal matrix of the non-zero singular values with square roots of the $X^T X$ matrix as elements and the residual E .

A common representation in chemometrics is the decomposition of the matrix X as the sum of the outer product of vectors score t_a and loadings p_a (Kumar et al. 2014).

$$X = t_1 p_1^T + t_2 p_2^T + \dots + t_A p_A^T + E \quad \text{or} \quad X = T P^T + E. \quad (6.71)$$

Here, the single difference from Equation (6.70) is the fact that V is identical to P but scaled to length one. A convenient way to generate the matrices T and P is based on non-linear iterative partial least square (NIPALS) applied to X , which is scaled to unit variance along each column. As described by Kumar et al. (2014), p_a^T can be defined as follows, with the score vector t_a selected from the column of matrix X that has the largest variance:

$$p_a^T = \frac{t_a^T X}{t_a^T t_a}. \quad (6.72)$$

Then, p_a^T can be normalised to unit length when multiplying by a constant factor c :

$$c = \frac{1}{\sqrt{p_a^T p_a}}. \quad (6.73)$$

The following equation is then used to find the new score vector:

$$t_a = \frac{X p_a}{p_a^T p_a}. \quad (6.74)$$

The new scores are then used to compute the loadings using Equation (6.72) and the procedure (Equations (6.72–6.74)) is repeated until convergence for the a th principal component. Subsequently, in order to compute the next principal component, the X matrix is deflated using $X = X - t_a p_a^T$. Equations (6.72–6.74) are applied using the deflated matrices to compute the subsequent principal components. Therefore, the residual E is iteratively decreased by adding new principal components along directions explaining largest variance.

Following one of the central goals of PCA, namely to simplify the visualisation and interpretation, in most of the cases, the high-dimensional data are then visualised in 2D plots of the lower-dimensioned scores to identify characteristic groups or of the lower-dimensioned loadings to study the characteristic interrelationship among the

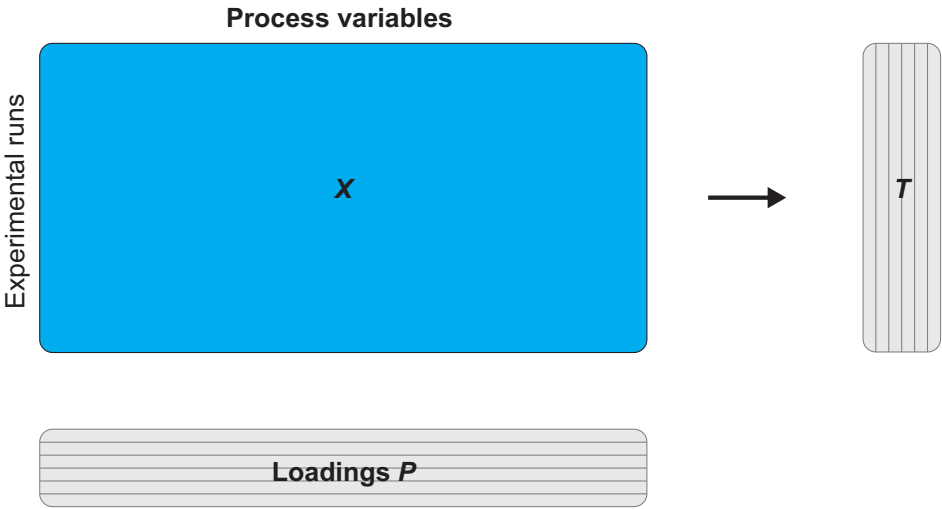
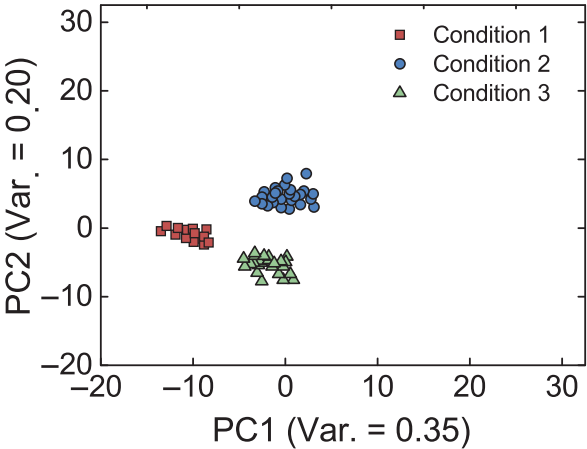


Figure 6.8 Simplified PCA on process data matrix X . The relation of the new system to the original one is given by the scores T (projection of the experiments) and the loadings P (directions of the original variables in the new space).

Figure 6.9 Exemplary score plot for the first two principal components for three process conditions (1, 2 and 3), the axes indicate the variance explained by the corresponding principal component. In this exemplary case, the resulting clusters allow the clear distinction of different process conditions in the experimental data set.



variables. An example of such a score plot is presented in Figure 6.9, where the first two PCs which jointly explain 55 per cent of the variability in the data are presented.

This data set was obtained in an analysis of the intracellular metabolite profile during sequential perfusion cultures. For this purpose, Karst et al. (2017c) measured highly dimensional MALDI-TOF MS spectra of intracellular metabolites for the sequence of three VCD set points (20×10^6 , 60×10^6 and 40×10^6 cells/mL) over 10 selected culture days. Applying PCA to these spectra revealed, as shown in Figure 6.9, a clear clustering of the three set points in the score plots of the first three principal components. Such cluster specific grouping can be used as an indication of a set-point specific intracellular metabolite profile and shows the visualisation potential of PCA. It can also be very supportive in identifying abnormal behaviour along the dynamic evolution of an experiment (Sokolov et al. 2015).

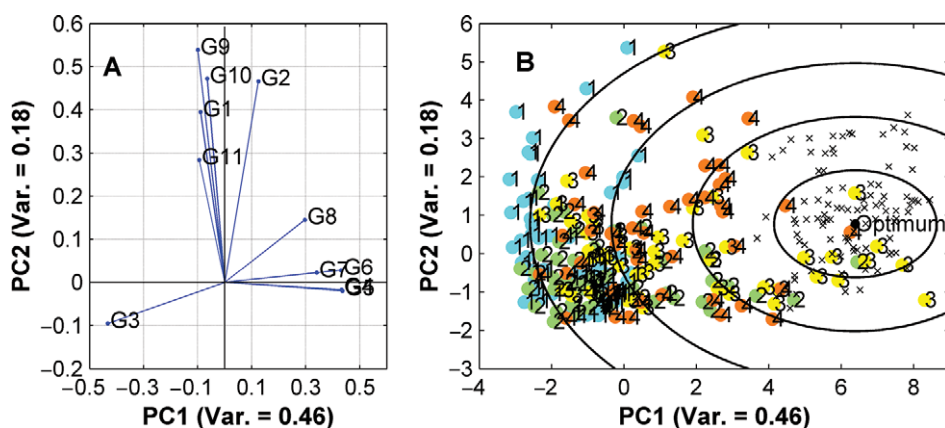


Figure 6.10 Y-PCA of glycoforms. (A) Loading plot of the 11 Y-variables; (B) the score plot is labelled according to the utilised clone and additionally shows the location of the optimum as well as of 100 projected corners of the specification range (cross markers). The ellipses visualise equidistant Mahalanobis distances from the optimum ranging from 1 to 4 U. ©Reprinted from *Biotechnology Progress*, Michael Sokolov, Jonathan Ritscher, Nicola MacKinnon, Jean-Marc Bielser, David Brühlmann, Dominik Rothenhäusler, Gian Thanei, Miroslav Soos, Matthieu Stettler, Jonathan Souquet, Hervé Broly, Massimo Morbidelli and Alessandro Buttè, ‘Robust factor selection in early cell culture process development for the production of a biosimilar monoclonal antibody’, 2016, with permission from John Wiley and Sons

The possibility to jointly interpret scores and loadings, which – given the mathematical definition – follow the same orientation, shall be demonstrated on a cell culture process development study for a biosimilar. Here, Sokolov et al. (2017b) applied PCA to 11 (scaled) final product quality variables shown in the loading plot in Figure 6.10A, where one can distinguish three groups of variables. These can be identified as three groups of lower, intermediate and higher processed glycoforms along the glycosylation network (see also Section 6.5.2). The score plot demonstrates the location of the final product quality of the performed experiments under different conditions with respect to the target product (highlighted by optimum and cross-marks for the specification range). Given the joint orientation, one can identify in a single experiment that to reach optimal biosimilarity, the quantity of the forms G5 to G8 has to be increased (large values along PC1), while G3 and the other glycoforms have to be decreased (small values along PC2).

Partial Least Square Regression Partial least squares (PLS) analysis represents another important multivariate technique used for multivariate regression of one (PLS1) or several (PLS2) dependent variables (e.g., final product quality) based on correlated independent variables (e.g., process variables). The multivariate nature of the model enables the reduction of the dimensions of the interrelationship between, e.g., the process conditions and the product characteristics, to simplify interpretation. Hence, unlike PCA, PLS is not a visualisation but a regression tool enabling the prediction of one or several output variables based on several input variables. The major preparation steps for PLS are definition of input/independent variables x , and output/dependent variables y , problem definition, data preparation including alignment and unfolding (as product quality measurements usually less frequent than process measurements), as well as data pretreatment to ensure comprehensible

6.6 Statistical and Hybrid Modelling

variable comparison (Sokolov et al. 2015). The set of X in bioprocessing can, e.g., comprise measurement data of several process variables (such as VCD, GLC, LAC, pH, DO) or spectral data, while Y contains the targeted response variables, such as final titre concentration or product quality patterns, or reference measurements when spectral data are considered.

The X and Y data matrices are mean centred and scaled before applying the PLS regression. PLS essentially finds a set of orthogonal latent variables that best predicts Y as well as best models the X matrix. The number of latent variables (LVs) are typically much smaller than the number of independent variables (X). PLS aims at finding two sets of coefficients or weights namely W^* and C to create a linear combination of the X and Y columns, respectively, such that these linear combination have a maximum covariance. These linear combinations are called the X -scores (denoted by T) and the Y -scores (denoted by U) computed as shown here:

$$T = XW^* \quad (6.75)$$

$$U = YC. \quad (6.76)$$

Inner relations of the PLS regression model link the X and Y matrices together while the outer relations shown here describe them individually:

$$X = T \times P^T + E \quad (6.77)$$

$$Y = U \times C^T + F, \quad (6.78)$$

with P^T being the loading matrix of the X space, C^T the loading matrix of the Y space, and E and F the residual matrices of the X and Y spaces, respectively. X -scores, or T , are also good predictors for Y variables, as seen in the following equation:

$$Y = T \times C^T + G, \quad (6.79)$$

Using Equations (6.75) and (6.79), this then becomes

$$Y = XW^*C^T + G = XB + G, \quad (6.80)$$

where

$$B = W^*C^T, \quad (6.81)$$

with B being the PLS regression coefficient and G the residual matrix. Similar to the PCA, a NIPALS algorithm is used to compute the T , U , C , P and W^* , as nicely illustrated by Wold et al. (2001).

The portions of the model that cannot be explained are called residuals, and their value is useful to understand the applicability of the model. A good model can be distinguished from a poor model because it will feature low residual values. Residual matrices can be used to recalculate PLS components after the first model, and this can be done until 99 per cent of the variance can be explained. The number of significant PLS components required for a calibration model can be estimated using cross-validation (Cao et al. 2010, García-Muñoz & Polizzi 2012, Indahl 2005).

When training a PLS model, it is important to avoid overfitting and include only as many latent variables as are required to describe general patterns affecting the entire data set and not only very small subsections of it, which would limit later

6 Mechanistic and Statistical Modelling of Bioprocesses

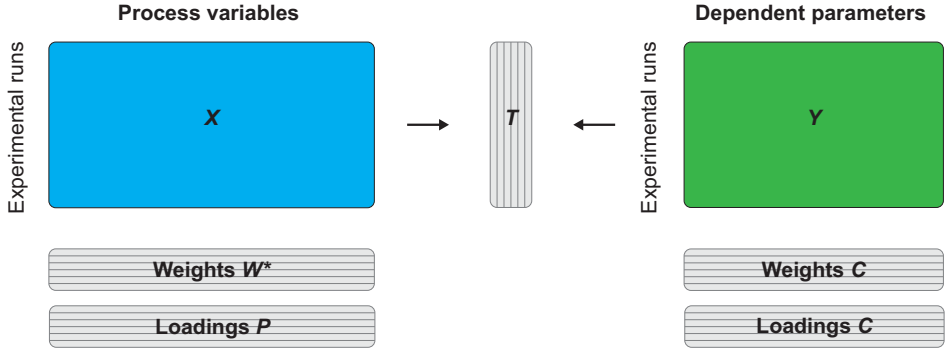


Figure 6.11 Simplified PLS from, e.g., process data matrix X to Y matrix containing product quality information. The high-dimensional relationship is reduced to a lower-dimensional number of latent variables, consisting of the scores T (projections of the experiments) and the corresponding loadings W^* and C (transformations of the original X and Y variables to the latent variable space).

its use to predicting new data. For this purpose, a technique called cross-validation is used, which iteratively trains models on subsets of the data and determines the optimal numbers of latent variables (Cao et al. 2010).

As a measure of model performance during calibration, the root mean square error in cross validation (RMSECV) can be defined as follows:

$$\text{RMSECV} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{\text{test},i} - \hat{y}_{\text{test},i})^2}. \quad (6.82)$$

with n representing the number of observations in the calibration set, which is in groups of K used as an internal test set so that each observation i is tested as $y_{\text{test},i}$ and estimated as $\hat{y}_{\text{test},i}$ by the corresponding model on the subset used for training. Such RMSECV is calculated for each additional LV, which results in an optimal number finding a trade-off between explaining the complex interrelationship between X and Y while not overfitting it. In addition, the relative variance explained in cross-validation (Q^2) can be considered, which is described as the ratio of explained (ESS) to total variance (TSS) in the internally predicted data.

$$Q^2 = 1 - \frac{\text{ESS}}{\text{TSS}}. \quad (6.83)$$

$Q^2 = 1$ signifies a perfect prediction model. This characteristic is maximal at the optimal number of LVs used. For a proper evaluation of the external prediction power, however, the best calibrated model has to be evaluated with an external data set, which was not part of the cross-validation procedure and is therefore completely unknown to the model. This step allows the testing of the robustness and extrapolation potential of the model towards new data. The root mean square error in prediction (RMSEP) is one of the most common performance indicators for this:

$$\text{RMSEP} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{\text{ext},i} - \hat{y}_{\text{ext},i})^2}, \quad (6.84)$$

6.6 Statistical and Hybrid Modelling

with n representing the number of observations in the corresponding external set, $y_{test,i}$ representing the y -value of the i th observation in the external set and $\hat{y}_{ext,i}$ representing the model estimation of the y -value of the i th observation.

PLS regression analysis represents a widely used technique in bioprocess data analysis. It has been used, for example, to predict and forecast final product quantity and quality attributes – such as titre, aggregates, fragments, charge variants and N-linked glycosylation in high-throughput cell culture experiments, across multiple other reactor scales (Sokolov et al. 2017c, 2018). Moreover, it is a widely used technique to build predictive models for relevant process variables (Y) based on spectra acquired, e.g., Raman or NIR sensors (Feidl et al. 2019).

As a detailed example, Sokolov et al. (2017c) investigated the results of 91 experimental fed-batch runs, all performed in the ambr15[®] system. The runs were based on four experimental blocks of DoEs with the objective to test nine different media supplements and three different process-execution strategies. The data were separated in three different characteristics blocks: Block Z , the experimental conditions, was spanned by the experimental runs (rows) and different process conditions (columns); Block X , the process dynamics, was built by the experimental runs (rows) measured process variables at different time points (columns) and Block Y represents the different final product quality attributes (columns) for the experimental runs (rows). Such separation exists not only to enable the learning of the characteristic interrelationship of the process to the product but also enables the identification of the Z parameters driving the process (which are important for process control) and X variables best representing the dynamic process behaviour (which are important for process monitoring). In the first step, a principal component analysis was used and revealed the strong correlation characteristics among the product quality attributes, similarly to Figure 6.10A. In the second step, partial least square regression was applied to predict product quality attributes, revealing the interrelationship of process characteristics and product quality variables. It shall be highlighted that here, given the strong identified correlation structure in Y , PLS2 models were used, linking the process to the product behaviour with a single model (instead of one model per product variable). This remarkably simplified the interpretation of the analysis results. Thereby, several models were built with different portions of process history incorporated – i.e., forecasting from the initial and designed conditions Z – as well as models incorporating some or all of the dynamic process history X .

Figure 6.12 presents the Q^2 obtained for such predictions for the different variables (columns) and different amounts of historic information used (rows). First of all, it shall be highlighted that the models can predict most of the quality variables very decently with Q^2 above 0.7. Even further, one can observe that LMW could be decently predicted based on using the initial conditions Z , while for variables such as the charge variant C4, at least information until culture day (T5) was required. This can be explained by the fact that several product quality attributes are allowing the prediction of these attributes early on in the process, while other attributes were more affected by process parameters, such as pH and temperature shifts, that were applied during the culture. This study highlights the strength of statistical models for large correlated data sets, to identify important dynamic patterns and process drivers to support decision making in process development.

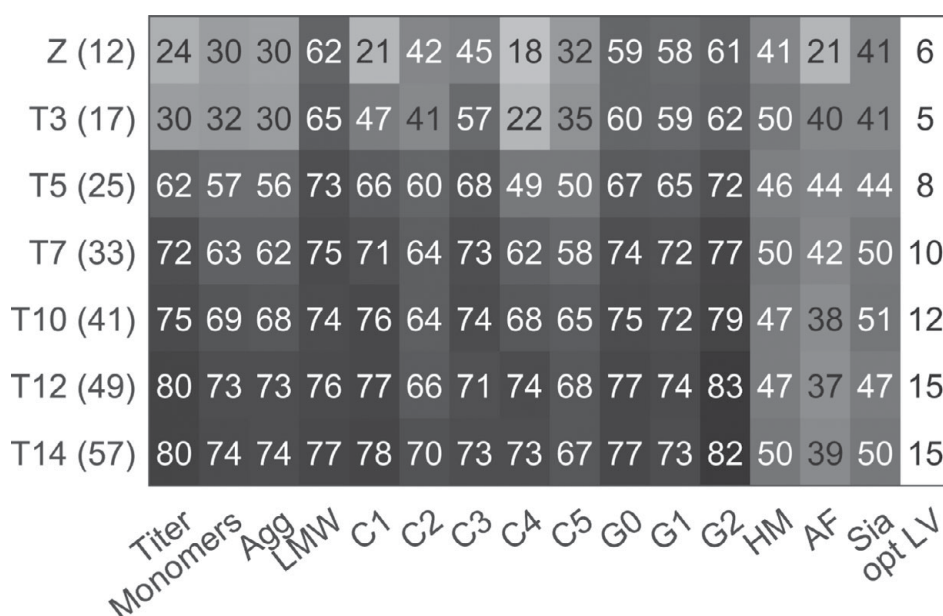


Figure 6.12 Variance explained in cross-validation (Q^2 in %) for PLS2 models for different CQAs at day 14 (columns) based on different lengths of process history utilised (rows, where the first one represents the controlled process conditions and the further ones refer to incorporating the complete process history). The numbers in brackets indicate the number of X variables used in the corresponding PLS2 model. © Reprinted from *Biotechnology Progress*, Michael Sokolov, Jonathan Ritscher, Nicola MacKinnon, Jonathan Souquet, Hervé Broly, Massimo Morbidelli and Alessandro Butté, ‘Enhanced process understanding and multivariate prediction of the relationship between cell culture process and monoclonal antibody quality’, 2017, with permission from John Wiley and Sons

6.7 HYBRID MODELLING APPROACH

Having shown the strengths of both modelling approaches, we have seen that while deterministic models require a sufficient understanding of the underlying mechanisms, statistical models require sufficiently large data sets to understand all the correlations for a given set of data. Hybrid modelling approaches represent a compromise of these two techniques by combining in a pragmatic way *a priori* knowledge of deterministic models as a structural backbone with the flexibility of a statistical model to describe unknown interrelationships inside that structure (Chen et al. 2004, Glassey & von Stosch 2018, Narayanan et al. 2019c, Von Stosch et al. 2016). A very common approach in hybrid modelling of bioprocesses is to apply statistical tools to estimate rate constants of a given deterministic reaction network or cell culture model, such as the cellular growth constant or specific metabolite production and consumption rates using either artificial neural networks or PLS models (Teixeira et al. 2005, Von Stosch et al. 2016).

An important example of hybrid modelling in bioprocessing was presented by Narayanan et al. (2019c). In this work, the added value of hybrid modelling compared to the alternative (particular statistical) approaches was presented from several perspectives. The analysis was performed on a cell culture process characterisation data set with almost 80 runs (Sokolov et al. 2015). As presented in Figure 6.13A, the hybrid modelling technique could significantly outperform statistical modelling

6.8 Advanced Process Monitoring and Control

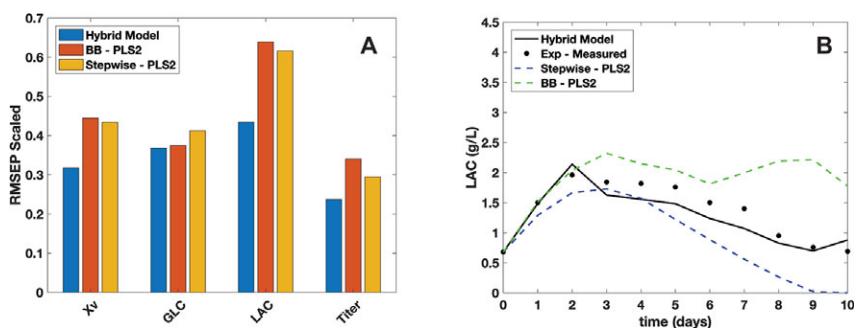


Figure 6.13 (A) Comparison of the accuracy of different models reported in terms of RMSEP scaled with respect to the standard deviation for X_v , glucose, lactate and titre and (B) shows the evolution of lactate for one run predicted by different models. © Reprinted from *Biotechnology and Bioengineering*, Harini Narayanan, Micheal Sokolov, Massimo Morbidelli and Alessandro Butté, ‘A new generation of predictive models: The added value of hybrid models for manufacturing processes of therapeutic proteins’, 2019, with permission from John Wiley and Sons

techniques to predict the complete evolution profiles of central variables such as lactate and titre. It shall be highlighted here that the goal of hybrid modelling in this analysis was to reliably predict the process behaviour on the designed variables only (Z variables). This shall enable one to define the process design space and reliable digital process twins to simulate process behaviour. The lack of available dynamic measurements is compensated by mechanistic process knowledge, while, for the statistical models, a day-wise propagation procedure is applied. Figure 6.13B demonstrates such enhanced predictive power for lactate. This predictive accuracy is essential to reduce the experimental effort and support experimental design in process development based on advanced models reflecting prior knowledge. Besides the potential to reduce the experimental design and more efficiently converge to an optimal design space, this study has additionally highlighted the important underlying capability for extrapolation clearly going beyond the limits of purely data-driven approaches.

6.8 ADVANCED PROCESS MONITORING AND CONTROL

After discussing the different types and applications of mathematical models, in this section, we elaborate on the use and application of these modelling techniques for process monitoring and control. The PAT/QbD initiative started in 2004 has resulted in worldwide increasing activity in the development of analytical and control strategies in bioprocesses (Glasse et al. 2011; Rathore 2014; Read et al. 2010a, 2010b). The recent advances in process automation and digitalisation technologies increasingly permeate the field of biotechnology, making experimental data from bioreactor runs more accessible and simpler to treat (Carrondo et al. 2012). The implementation of online spectral sensors, such as Raman or NIR, measuring thousands of wavelengths every few minutes, also produce significant amounts of additional data, thus providing new tools for advanced process monitoring and control.

We have seen in the previous section that multivariate data analysis methods are perfectly suited for data processing, reducing data dimensions and noise disturbance, identifying important patterns and trends as well as constructing predictive models. Thereby, statistical regression models or even models based on advanced mechanistic modelling could be used for developing soft sensors based on online Raman or NIR

spectra (Feidl et al. 2019, Sommeregger et al. 2017). The spectra from biological samples are, in fact, complex and diverse, and interpreting them to extract the desired information is difficult. Therefore, after proper data preprocessing – which is not discussed in detail here, although it is an important step to be executed with great care – PCA and PLSR analysis can be used to build predictive models that can potentially enable the monitoring of various process variables ranging from measurable species – such as VCD, GLC or mAb concentration – to non-measurable species, such as critical product quality attributes (Escandar et al. 2006, Gautam et al. 2015, Hubert & Engelen 2004). In addition, the application of these tools combined with tools enabling process modelling and optimisation, such as the hybrid model presented earlier, can enable the proactive identification of risks of process failure much earlier – e.g., when certain process parameters start to move out of the process design space – and thus allow the operator to take proper actions.

It appears that such model-based techniques can cover a versatile set of supportive tasks ranging from descriptive (dimension reduction and trend monitoring), diagnostic (alert systems and failure analysis), predictive (soft sensors, forecasting models) and prescriptive (process control and optimisation) applications (Narayanan et al. 2019a). With regards to the ultimate goal in biopharmaceutical manufacturing, namely that of consistently providing product quality within specifications, these tools can be further improved so as to implement a direct control of the critical quality attributes. Such implementation of predictive quality coupled with process control shall eventually advance towards adaptive solutions, shaping the path towards real-time release or at least release by acceptance. Besides data analytics and process modelling tools, data management and integration are crucial to realising the potential of process digitalisation and automation (Luttmann et al. 2015).

The current design of manufacture facilities uses either a *modular approach*, where each unit operation is considered as a separate independent unit that is optimised independently and then integrated with another module without any modification, or an *adaptation approach*, where one- or two-unit operations to be integrated are adapted in a manner that allows their seamless integration. This situation is fundamentally changed in integrated continuous biomanufacturing, where the entire manufacturing facility is basically optimised as a single large unit, thus taking advantage of all the interconnections amongst the single units to design their individual operations to optimise the behaviour of the entire manufacturing facility. This implies overcoming the traditional separation between upstream and downstream units and the collection and elaboration of the data coming from all sensors on different units in a single place, as illustrated in the schematic of a continuous integrated unit shown in Figure 6.14 (Konstantinov & Cooney 2015, Steinebach et al. 2017). For example, the performance of the capture step in a mAb production facility is significantly affected by the behaviour of the perfusion bioreactor. Accordingly, the integration of these two units can be most beneficial (Karst et al. 2017a). For this, it is necessary to monitor the output of the perfusion bioreactor and use it to control the capture step. The bioreactor is, in fact, more prone to deviations in its behaviour as a function of time due to inherent complexities of the culture process, such as cell ageing or metabolic changes leading to decreasing titre values. In such cases, real-time monitoring of product concentration and quality is necessary to adapt the operating conditions of the capture unit in order to keep the desired overall performance. This has to be done at two levels: the highest one, where the product has to be kept within specifications (otherwise, it will have to be discarded), and a low one, where the cost

6.9 Conclusion

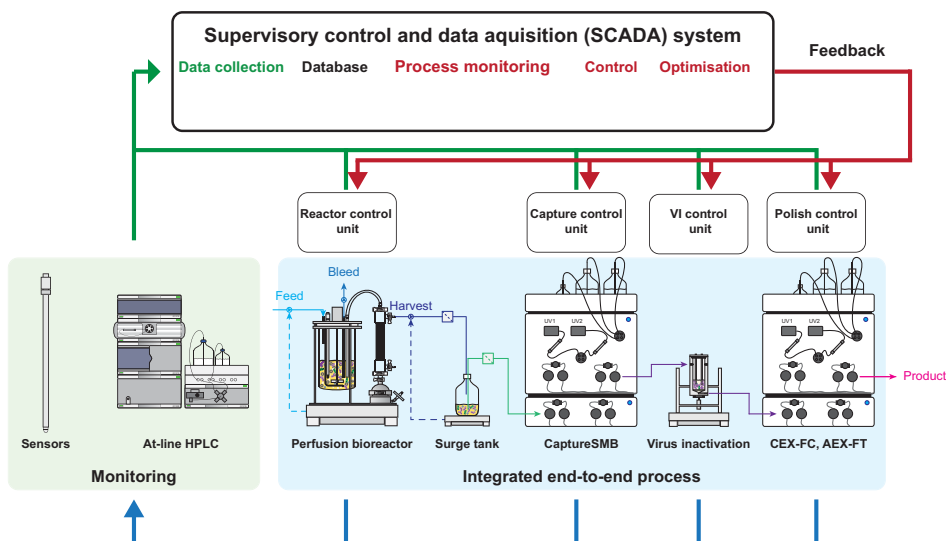


Figure 6.14 Schematic representation of an integrated continuous process with the implementation of sensors and at-line HPLCs for process monitoring. The local control loops of each unit (bioreactor, capture, polish, analytics, spectral data) are connected to a supervisory control and data acquisition system that collects all the data and applies proper data processing steps, in order to perform process monitoring, control and optimisation

of the operation has to be minimised, such as reducing the amount of buffer used per unit of mAb produced. An example of this has been provided by Karst et al. (2017a) where an at-line HPLCs (ProtA, SEC) was used to monitor concentration of the target protein and of its aggregates. This same concept can, in principle, be realised at higher level using the online spectroscopic sensors described earlier (Feidl et al. 2019). The next step is to integrate the available technologies into the entire continuous operation train and to build up control and monitoring solutions for the entire process, as illustrated in Figure 6.14 (Feidl et al. 2020).

6.9 CONCLUSION

In this chapter, we have shown the diversity of different quantitative modelling approaches that can be applied to cell culture processes. These cover general models describing stirred tank bioreactors, fed-batch or perfusion, including unstructured models for describing the cell culture growth and metabolism. These can be coupled with more specific models to obtain tools for controlling relevant and difficult details of the process, like the final product quality attributes. An example of this is the introduction of a dynamic mechanistic model of N-linked glycosylation to control and modulate the mAb glycosylation patterns. Moreover, we discussed the use of statistical and hybrid models enabling the analysis of large data sets and the implementation of soft sensors. With respect to the current trends, model-based advanced process monitoring, optimisation, control and automation techniques will represent key enablers of continuous biomanufacturing, as they enable the reduction of the experimental effort and facilitate the integration, operation and understanding of the different and complex process units.